# Class 10

# Mathematics

## Volume 2

## Solutions

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# 4. Indefinite Integrals

## WORKSHEET – 1

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) 3 | 2) 2 | 3) 1 | 4) 2 | 5)1 | 6) 2 | 7) 1 |
| 8)1 | 9)1 | 10)2 | 11)1 | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) A | 2) A | 3) B | 4) B | 5) D | 6) A | 7) B |
| 8) A | 9) B | 10) A | 11) B | 12) C | 13) A | 14) A |
| 15) A | 16) C | 17) B | 18) A | 19) A | 20) C | 21) C |
| 22) A,B,C | 23) A | 24) B | 25) B | 26) B | | |

27) $a \to r; b \to p; c \to q; d \to s$ 28)B 29)8 30) $a \to r; b \to s; c \to p; d \to q$

31)A 32) $a \to r; b \to s; c \to p; d \to q$

1. $x^4 + x^2 + 1 = (x^2)^2 + 2x^2 + 1 - x^2 = (x^2 + 1)^2 - x^2 = (1 + x^2 + x)(x^2 - x + 1)$
2. Multiply numerator and denominator by $1 - \sin x$
3. $\cos 2x = 1 - 2\sin^2 x, \cos 2\alpha = 1 - 2\sin^2 \alpha$
4. $\cot^2 x = \operatorname{cosec} 2x - 1$
5. $\cosh x + \sinh x = e^x$
6. $\int (2^x + 2^{-x})^2 \, dx = \int (2^{2x} + 2^{-2x} + 2)dx = \int (4^x + 4^{-x} + 2)dx = \frac{4^x}{\log 4} - \frac{4^{-x}}{\log 4} + 2x + c$
7. $\int \frac{1-\cos 2x}{1+\cos 2x} dx = \int \frac{2\sin^2 x}{2\cos^2 x} dx = \int \tan^2 x dx = \int (\sec^2 x - 1)dx = \tan x - x + c$
8. $\int \frac{1}{\sqrt{x^2+2x+1}} dx = \int \frac{1}{x+1} dx (x < -1)\Big) = \log|x + 1| + c; k = 1$
9. $\frac{x^4}{x+x^5} = \frac{(x^4+1)-1}{x(x^4+1)}$
10. $1 - \sqrt{x} = (1 + \sqrt[4]{x})(1 - \sqrt[4]{x})$
11. $\sec x - \tan x = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$
12. Multiply numerator and denominator by $\sqrt{x + a} - \sqrt{x + b}$
13. Write $e^x = e^{x/2} \cdot e^{x/2}$ and then put $e^{x/2} - 1 = t$
14. $\int \sqrt{1 + \sin 2x} dx = \int (\sin x + \cos x)dx = -\cos x + \sin x + c$
15. $\sinh x = \frac{e^x - e^{-x}}{2}$

16. $\int \frac{8^{1+x}+4^{1+x}}{2^x}dx = \int \left(\frac{8.8^x}{2^x}+\frac{4.4^x}{2^x}\right)dx = 8\int (4)^x dx + 4\int 2^x dx = \frac{8.4^x}{\log 4}+\frac{4\times 2^x}{\log 2}+c$

17. $\int \frac{x^5}{x^2+1}dx = \int \left[x^3 - x + \frac{x}{x^2+1}\right]dx = \frac{x^4}{4}-\frac{x^2}{2}+\frac{1}{2}\log(x^2+1)+c$

18. $\int \frac{\sqrt{1-x^2}+\sqrt{1+x^2}}{\sqrt{1-x^4}}dx = \int \left(\frac{1}{\sqrt{1+x^2}}+\frac{1}{\sqrt{1-x^2}}\right)dx = \log\left(x+\sqrt{x^2+1}\right)+\sin^{-1}x+c.$

19. $\int \cos^3 x dx = \int \left(\frac{\cos 3x+3\cos x}{4}\right)dx = \frac{1}{4}\frac{\sin 3x}{3}+\frac{3}{4}\sin x + c$

20. $\int \cos 4x\cos 3x dx = \frac{1}{2}\int 2\cos 4x\cos 3x dx = \frac{1}{2}\int [\cos(4x+3x)+\cos(4x-3x)]dx$
$= \frac{1}{2}\int \cos 7x dx + \frac{1}{2}\int \cos x dx = \frac{1}{2}\left[\frac{\sin 7x}{7}\right]+\frac{1}{2}(\sin x)+c = \frac{1}{2}\left[\frac{\sin 7x}{7}+\sin x\right]+c$

21. $\int \sqrt{1+\sin x}dx = \int \sqrt{\cos^2\frac{x}{2}+\sin^2\frac{x}{2}+2\cos\frac{x}{2}\sin\frac{x}{2}} = \int \sqrt{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2}dx =$
$\int \left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)dx = 2\sin\frac{x}{2}-2\cos\frac{x}{2}+c$

22. Solving $\int 2x^2 dx = 2\int x^2 dx = 2\times\frac{x^3}{3} = \frac{2x^3}{3}$

23. Both the statements are true.

24. $\int \frac{\sin^3 x}{\sin^2 x\cos^2 x}dx = \int \frac{\sin x}{\cos^2 x}dx = \int \sec x\tan x = \sec x + c$

25. $\int \frac{(2x+3)(3x-2)}{x^2}dx = \int \left(\frac{6x^2-4x+9x-6}{x^2}\right)dx = \int \frac{6x^2+5x-6}{x^2}dx$

$= \int \left(6+\frac{5}{x}-\frac{6}{x^2}\right)dx = 6\int 1dx + 5\int \frac{1}{x}dx + 6\int -\frac{1}{x^2}dx = 6x+5\log|x|+\frac{6}{x}+c$

26. $\int \frac{\sin^2 x}{1+\cos x}dx = \int \frac{1-\cos^2 x}{1+\cos x} = \int \frac{(1+\cos x)(1-\cos x)}{1+\cos x}dx$
$= \int (1-\cos x)dx = \int 1dx - \int \cos x dx = x - \sin x + c$

27. a) $\int \sqrt[5]{x^3}dx = \int x^{3/5}dx = \frac{5}{8}x^{8/5}+c$

b) $\int (\sin x + \cos x - \sec^2 x)dx = -\cos x + \sin x - \tan x + c$

c) $\int \left(3e^x + 4\cdot 2^x + \frac{7}{x}\right)dx = 3e^x + 4\cdot\frac{2^x}{\log 2}+7\log|x|+c$ d) $\int \frac{1}{3x^2}dx = -\frac{1}{3x}+c$

28. $\int 10x^4 dx = 10\int x^4 dx = 10\times\frac{x^5}{5} = 2x^5 + C$

29. $\int \sin 5x\cos 3x dx = \frac{1}{2}\int 2\sin 5x\cos 3x dx = \frac{1}{2}\int [\sin(5x+3x)+\sin(5x-3x)]dx =$
$\frac{1}{2}\int \sin 8x dx + \frac{1}{2}\int \sin 2x dx = \frac{1}{2}\left[\frac{-\cos 8x}{8}+\frac{-\cos 2x}{2}\right]+c = \frac{1}{2}\left[\frac{-\cos 2x}{2}-\frac{1}{8}\cos 8x\right]+c$

30. a) $\int \sqrt{1+\cos 2x}dx = \int \sqrt{2\cos^2 x}dx = \sqrt{2}\int \cos x dx = \sqrt{2}\sin x + c$

b ) $\int \sqrt{1-\cos 2x}dx = \int \sqrt{2\sin^2 x}dx = \sqrt{2}\int \sin x dx = -\sqrt{2}\cos x + c$
c) $\int \sqrt{1+\sin 2x}dx = \int (\sin x + \cos x)dx = -\cos x + \sin x + c$

d) $\int \sqrt{1-\sin 2x}dx = \int (-\sin x + \cos x)dx = \sin x + \cos x + c$

31. $\int \frac{\sin^8 x - \cos^8 x}{1-2\sin^2 x\cos^2 x}dx = \int \frac{(\sin^4 x+\cos^4 x)(\sin^4 x-\cos^4 x)}{(\sin^4 x+\cos^4 x)}dx$

$\int (\sin^4 \mathrm{x} - \cos^4 \mathrm{x})\mathrm{dx} = \int (\sin^2 \mathrm{x} - \cos^2 \mathrm{x})\mathrm{dx} = \int -\cos 2\mathrm{x}dx = -\frac{1}{2}\sin 2\mathrm{x} + \mathrm{c}$

$\Rightarrow \mathrm{A} = -\frac{1}{2}$

32. a) $\int \frac{1}{1+\cos x}dx = \int \frac{1-\cos x}{\sin^2 x}dx = -\cot x + \operatorname{cosec} x + c$

b) $\int \frac{1}{1-\cos x}dx = \int \frac{1+\cos x}{\sin^2 x}dx = -\cot x - \operatorname{cosec} x + c$

c) $\int \frac{1}{1-\sin x}dx = \int \frac{1+\sin x}{\cos^2 x}dx = \tan x + \sec x + c$

d) $\int \frac{1}{1+\sin x}dx = \int \frac{1-\sin x}{\cos^2 x}dx = \tan x - \sec x + c$

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## WORKSHEET – 2

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1)A | 2) B | 3) D | 4) C | 5) A | 6)B | 7)B |
| 8)C | 9)D | 10)D | | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1)C | 2)A | 3)A | 4)B | 5)A | 6)B | 7)D |
| 8)A | 9)B | 10)B | 11)C | 12)D | 13)A | 14)B |
| 15)A | 16)B | 17)D | 18)C | 19)A | 20)A | 21)B |
| 22) A,C,D | 23) A | 24) A | 25) B | 26) A | | |
| 27) $a \to r; b \to p; c \to q; d \to s$ | | | 28) B | 29) B,D | | |
| 30) $a \to r; b \to s; c \to p; d \to q$ | | | 31)C | 32)B | 33)C | |
| 34) $a \to r; b \to s; c \to q; d \to p$ | | | 35)A | | | |

1. Let $\mathrm{a}^2 - \mathrm{x}^3 = \mathrm{t} \Rightarrow -3\mathrm{x}^2\mathrm{dx} = \mathrm{dt} \Rightarrow \mathrm{x}^2\mathrm{dx} = -\frac{1}{3}\mathrm{dt}$

$\therefore \int \frac{-\frac{1}{3}\mathrm{dt}}{\sqrt{\mathrm{t}}} = -\frac{1}{3}\int \frac{1}{\sqrt{\mathrm{t}}}\mathrm{dt} = -\frac{1}{3}(2\sqrt{\mathrm{t}}) + \mathrm{c} = \frac{-2}{3}\sqrt{\mathrm{a}^2 - \mathrm{x}^3} + \mathrm{c}$

2. Let $\mathrm{e}^{\mathrm{x}} + \mathrm{e}^{-\mathrm{x}} = \mathrm{t} \Rightarrow (\mathrm{e}^{\mathrm{x}} - \mathrm{e}^{-\mathrm{x}})\mathrm{dx} = \mathrm{dt} \ \therefore \int \frac{1}{\mathrm{t}}\mathrm{dt} = \log|\mathrm{t}| = \log|\mathrm{e}^{\mathrm{x}} + \mathrm{e}^{-\mathrm{x}}| + \mathrm{c}$

3. Let $2 + 3\cos^2 \mathrm{x} = \mathrm{t} \Rightarrow -3(2\sin \mathrm{x}\cos \mathrm{x})\mathrm{dx} = \mathrm{dt} \Rightarrow \sin 2\mathrm{x}\mathrm{dx} = -\frac{1}{3}\mathrm{dt}$

$\therefore \int \frac{-\frac{1}{3}\mathrm{dt}}{\mathrm{t}} = -\frac{1}{3}\int \frac{1}{\mathrm{t}}\mathrm{dt} = -\frac{1}{3}\log|\mathrm{t}| + \mathrm{c}$

4. Let $\mathrm{Tan}^{-1}\mathrm{x} = \mathrm{t} \Rightarrow \frac{1}{1+\mathrm{x}^2}\mathrm{dx} = \mathrm{dt} \ \therefore \int \frac{1}{\mathrm{t}}\mathrm{dt} = \log|\mathrm{t}| + \mathrm{c} = \log|\mathrm{Tan}^{-1}\mathrm{x}| + \mathrm{c}$

5. Put $\cos x = t$

6. Apply the formula $\int (f(x))^n f^1(x)dx = \frac{[f(x)]^{n+1}}{n+1} + c$

7. Put $5x^7 = t$

8. $\int \sum_{r=0}^{\infty} \frac{x^r 3^r}{r!} dx = \int \left(1 + \frac{3x}{1!} + \frac{3^2x^2}{2!} + \cdots \ldots\right) dx$

9. $\int \frac{1}{\sqrt{2ax-x^2}} dx = \int \frac{1}{\sqrt{2ax-x^2+a^2-a^2}} dx = \int \frac{1}{\sqrt{a^2+x^2-2ax-a^2}} = \int \frac{1}{\sqrt{(a-x)^2-a^2}} dx = \sin^{-1}\left(\frac{x-a}{a}\right) + c.$

10. $\int 3^{-\log_9 x^2} dx = \int \frac{1}{x} dx$

11. $\tan(A-B) = \frac{\tan A - \tan B}{1+\tan A \tan B}$

12. Put $\sin^{-1}(x^2) = t$

13. G.I. $= \frac{1-\cos x}{1-\cos^2 x} = \frac{1}{1+\cos x}$

14. Put $xe^x = t$

15. Put $e^x - 1 = t^4$

16. $\int \frac{2x}{\sqrt{1-4^x}} dx = \int \frac{2x}{\sqrt{1-(2^x)^2}} dx$ put $2^x = t$ 17. Put $\tan^{-1}\sqrt{x} = t$

17. $(x+\alpha) - \alpha = x$. Taking tan on both sides 19. Put $\cos^{-1} 3x = t$

18. Multiply numerator and denominator by $1 - \sin ax$

19. $f(x) = \frac{\cos x}{1+\cos x} = \frac{(1+\cos x)-1}{1+\cos x}$ Split and integrate

20. Let $\tan x = t \Rightarrow \sec^2 x dx = dt; \int \sec^2 x \operatorname{cosec}^4 x dx = \int \sec^2 x(1+\cot^2 x)^2 x dx = \int \left(1+\frac{1}{t^2}\right)^2 dt$

$$\int \left(1+\frac{1}{t^2}\right)^2 dt = \int (1+2t^{-2}+t^{-4})dt = t - \frac{2}{t} - \frac{1}{3t^3} + c = \tan x - 2\cot x - \frac{1}{3}\cot^3 x + c$$

$$\therefore k = \frac{-1}{3}; L = 1; \mathrm{M} = -2; \mathrm{K} + \mathrm{L} + \mathrm{M} = \frac{-4}{3}$$

23. Both the statements are true 24. Conceptual.

24. $\int \frac{\mathrm{dx}}{\sqrt{9-25\mathrm{x}^2}} = \frac{1}{5}\int \frac{\mathrm{dx}}{\sqrt{\frac{9}{25}-\mathrm{x}^2}} = \frac{1}{5}\sin^{-1}\left(\frac{\mathrm{x}}{\left(\frac{3}{5}\right)}\right) = \frac{1}{5}\sin^{-1}\left(\frac{5\mathrm{x}}{3}\right) + \mathrm{c}$

25. Multiplying the number and denominator by $e^{-x}$, we get

$$\int \frac{\mathrm{dx}}{\sqrt{1-\mathrm{e}^{2\mathrm{x}}}} = \int \frac{\mathrm{e}^{-\mathrm{x}}}{\sqrt{\mathrm{e}^{-2\mathrm{x}}(1-\mathrm{e}^{2\mathrm{x}})}} = \int \frac{\mathrm{e}^{-\mathrm{x}}}{\sqrt{\mathrm{e}^{-2\mathrm{x}}-1}}$$

Putting $e^{-x} = t = -\log\left|t + \sqrt{t^2-1}\right| + c = -\log\left|e^{-x} + \sqrt{e^{-2x}-1}\right| + c$

27. a) $\int \frac{1}{\sqrt{4-x^2}} dx = \int \frac{1}{\sqrt{2^2-x^2}} dx = \sin^{-1}\left(\frac{x}{2}\right) + c$

b) $\int \frac{1}{\sqrt{4+x^2}} dx = \int \frac{1}{\sqrt{2^2+x^2}} dx = \sinh^{-1}\left(\frac{x}{2}\right) + c$

c) $\int \frac{1}{\sqrt{x^2-4}}dx = \int \frac{1}{\sqrt{x^2-2^2}}dx = \cosh^{-1}\left(\frac{x}{2}\right) + c$ d) $\int \frac{1}{\sqrt{9-x^2}}dx = \sinh - 1\left(\frac{x}{3}\right) + c$

28. Let $3 - 2x = t \Rightarrow -2dx = dt \Rightarrow dx = \frac{-1}{2}dt \ \therefore \frac{-1}{2}\int \frac{1}{t}dt = -\frac{1}{2}\log|t| = \frac{-1}{2}\log|3-2x|$

29. Putting $x^{\frac{3}{2}} = t$, we have $\sqrt{x}dx = \frac{2}{3}dt$

So $\int \frac{x^{\frac{1}{2}}}{\sqrt{1-x^3}}dx = \frac{2}{3}\int \frac{dt}{\sqrt{1-t^2}} = \frac{2}{3}\sin^{-1}t + c = \frac{2}{3}\sin^{-1}x^{\frac{3}{2}} + c$

Thus $f(x) = x^{\frac{3}{2}}$ and $g(x) = \sin^{-1}x$.

30. a) Take $\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt \Rightarrow \frac{1}{\sqrt{x}}dx = 2dt \Rightarrow 2\int \sin t\,dt = -2\cos t + c = -2\cos(\sqrt{x}) + c.$

b) Take $\sqrt{x} = t$ c) Take $\sqrt{x} = t$.

d) Take $\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt \Rightarrow \frac{1}{\sqrt{x}}dx = 2dt \Rightarrow 2\int a^t dt = \frac{2a^t}{\log a} = \frac{2}{\log a}a^{\sqrt{x}} + c.$

31. $\int \sin 4x\cos 3x dx = \frac{1}{2}\int 2\sin 4x\cos 3x dx = \frac{1}{2}\int (\sin 7x + \sin x)dx$ and apply concept.

32. Put $\sin x = t$. 33. Put $t = \log(\sec x + \tan x)$.

33. $a \to r; b \to s; c \to q; d \to p$. 35. Both the statements are true.

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## WORKSHEET – 3

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) A | 2) B | 3) C | 4) A | 5) A | 6) B | 7) B |
| 8) C | 9) A | 10) B | | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) A | 2) C | 3) B | 4) B | 5) C | 6) D | 7) C |
| 8) A | 9) B | 10) A | 11) C | 12) D | 13) A | 14) A |
| 15) B | 16) A | 17) A | 18) A | 19) 8 | 20) A | 21) B |
| 22) C | 23) 5 | 24) A,C | 25) A | 26) B | 27) B | |
| 28) a-r; b-s; c-p; d-q | | 29) 5 | 30) C | 31) a-q; b-p; c-r; d-s | | 32) A |
| 33) A | 34) A | 35) a-t; b-p,t; c-q; d-r | | | | |

1. Use $\int e^x u dx$
2. Put $U = x^2, V = 2^{3x}$
3. Put $I = \int \frac{\cos x}{x}dx$ and apply byparts
4. Put $x^2 = t$
5. $\int e^x(f(x) + f^1(x))dx = e^x f(x) + c$

6. Apply the formula $\mathrm{I_n} = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$

7. Apply the formula $\mathrm{I_n} = \frac{-\cot^{n-1} x}{n-1} - I_{n-2}$

8. $\int e^x \cos^3 x dx = \int e^x \left[\frac{\cos 3x + 3\cos x}{4}\right] dx = \frac{1}{4}\int e^x \cos 3x + \frac{3}{4}\int e^x \cos x dx$

$$= \frac{1}{4}\left[\frac{e^x}{1^2+3^2}(\cos 3x + 3\sin 3x)\right] + \frac{3}{4}\left[\frac{e^x}{1^2+1^2}(\cos x + \sin x)\right] + c$$

$$= \frac{e^x}{40}(\cos 3x + 3\sin 3x) + \frac{3e^x}{8}(\cos x + \sin x) + c$$

9. $\int x^3 e^{-x} dx = x^3 \int e^{-x} dx - \int 3x^2 \int e^{-x} dx \Rightarrow -x^3 e^{-x} + 3\int x^2 e^{-x} dx \Rightarrow -e^{-x}[x^3 + 3x^2 + 6x + 6] + c$

10. $x + 4 = (x + 6) - 2$

11. $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$

12. $\int e^x 2^{3\log_2 x} dx = \int e^x \cdot x^3 dx = x^3 \int e^x dx - \int 3x^2 \int e^x dx = e^x[x^3 - 3x^2 + 6x - 6] + c$

13. Put $\sqrt{x} = t$ 14. $I_5 = \int \mathrm{cosec}^5 x dx = \frac{-\mathrm{cosec}\, e^3 x \cot x}{4} + \frac{3}{4} I_3$

14. $\frac{1-\sin x}{1-\cos x} = \frac{1}{2}\mathrm{cosec}\, e^2 \frac{x}{2} - \cot \frac{x}{2}$

15. Apply the formula $\int e^x [f(x) + f^1(x)] dx$

16. Use $\int_e^{kx} [kf(x) + f'(x)] dx$ 18. Put $\mathrm{U = x^n, V = e^{cx}}$

17. $I = \int e^{3x} \sin 5x dx = \int e^{3x} \frac{d}{dx}\left(-\frac{1}{5}\cos 5x\right)$ and use integration by parts

18. $x^3 + x + 1 = x(1 + x^2) + 1$ 21. $t = \log_e x, I = \int e^t \left(\frac{1}{t} - \frac{1}{t^2}\right) dt$

19. $\int \frac{4\sec^2 x \tan x}{1 + 2\tan^2 x} dx$ put $2\tan^2 x = t$

20. $\int 3^x \cos x dx = \int \cos x \left(\frac{d}{dx}\left(\frac{3^x}{\log 3}\right)\right) dx$ and use integration by parts

21. $\int e^x \left(\frac{\cos x + \sin x}{1 + \cos 2x}\right) dx = \int e^x \left[\frac{\cos x}{2\cos^2 x} + \frac{\sin x}{2\cos^2 x}\right] dx = \frac{1}{2}\int e^x [\sec x + \tan x \sec x] dx = \frac{1}{2}\mathrm{e^x \sec x + c}$

22. We have, $I_n = \int \tan^n x dx = \int \tan^{n-2} x \cdot \tan^2 x dx = \frac{\tan^{n-1}}{n-1} - I_{n-2}$

$$\therefore I_{n+} I_{n-2} = \frac{\tan^{n-1} x}{n-1}, n \geq 2 = I_0 + I_1 + 2(I_2 + I_3 + \cdots .. + I_8) + I_9 + I_{10}$$
$$= (I_2 + I_0) + (I_3 + I_1) + (I_4 + I_2) + (I_5 + I_3) + (I_6 + I_4)$$
$$+(I_9 + I_5) + (I_8 + I_6) + (I_9 + I_6) + (I_9 + I_7) + (I_{10} + I_8) = \left(\frac{\tan x}{1} + \frac{\tan^2 x}{2} + \cdots . + \frac{\tan^9 x}{9}\right).$$

26. $I_n - I_{n-2} = \int \frac{\cos nx - \cos(n-2)x}{\sin x} dx = \int \frac{-2\sin(n-1)x \sin x}{\sin x} dx = \frac{2\cos(n-1)x}{n-1}$.

27. Let $I_{m,n} = \int \frac{\sin^m x}{\cos^n x}dx = \int \sin^{m-1} x \cdot \frac{\sin x}{\cos^n x}dx$

Integrating by parts, we have

$$I_{m,n} = \sin^{m-1} x \frac{1}{(n-1)\cos^{n-1} x} - \frac{m-1}{n-1}\int \frac{\sin^{m-2} x \cos x}{\cos^{n-1} x}dx$$
$$= \frac{\sin^{m-1} x}{(n-1)\cos^{n-1} x} - \frac{m-1}{n-1}\int \frac{\sin^{m-2} x}{\cos^{n-2} x}dx = \frac{\sin^{m-1} x}{(n-1)\cos^{n-1} x} - \frac{m-1}{n-1} I_{m-2,n-2}$$

28. a) $\int e^x(\sin x + \cos x)dx = e^x \sin x + c$ b) $\int e^x(\cos x - \sin x)dx = e^x \cos x + c$

c) $\int e^x(\tan x + \sec^2 x)dx = e^x \tan x + c$

d) $\int e^x(\tan x + \log \sec x)dx = e^x(\log \sec x) + c$

29. $\int e^x \cos 2x dx = \frac{e^x}{1+4}[\cos 2x + 2\sin 2x] = \frac{e^x}{5}[\cos 2x + 2\sin 2x]$

30. We have $I_n = \int \tan^n x dx = \int \tan^{n-2} x \cdot \tan^2 x dx$

$$= \int \tan^{n-2} x(\sec^2 x - 1)dx = \int \tan^{n-2} x \sec^2 x dx - \int \tan^{n-2} x dx = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$
$$\therefore I_n + I_{n-2} = \frac{\tan^{n-1} x}{n-1}, n \geq 2$$
$$\therefore I_0 + I_1 + 2(I_2 + I_3 + \cdots + I_8) + I_9 + I_{10} = (I_2 + I_0) + (I_3 + I_1) + (I_4 + I_2) + (I_5 + I_3)$$
$$+(I_6 + I_4) + (I_7 + I_5) + (I_8 + I_6) + (I_9 + I_7) + (I_{10} + I_8)$$
$$= \left(\frac{\tan x}{1} + \frac{\tan^2 x}{2} + \cdots + \frac{\tan^9 x}{9}\right)$$

32. $I_{n,m} = \int \underset{II}{x^{n-1}} \underset{I}{(1-x)^m} dx = \frac{(1-x)^m \cdot x^n}{n} + m\int (1-x)^{m-1} \frac{x^n}{n} dx = \frac{(1-x)^m \cdot x^n}{n} + \frac{m}{n} I_{n+1,m-1}$

33. $I_n = \int \frac{dx}{(x^2+a^2)^n} = \int \underset{II}{1} \underset{I}{\frac{(x^2+a^2)^{-n}}{}} dx$

$$I_n = (x^2 + a^2)^{-n} x + n\int 2x(x^2 + a^2)^{-n-1} x dx = x(x^2 + a^2)^{-n} + 2n\int (x^2 + a^2 - a^2)(x^2 + a^2)^{-n-1} dx$$
$$= x(x^2 + a^2)^{-n} + 2n\left[\int (x^2 + a^2)^{-n} - a^2 \int (x^2 + a^2)^{-n-1}\right]$$
$$I_n = x(x^2 + a^2)^{-n} + 2nI_n - 2na^2 I_{n+1}; (1-2n)I_n = x(x^2 + a^2)^{-n} - 2na^2 I_{n+1}$$

Replace $n \to n-1$

$$(-2n+3)I_n = x(x^2 + a^2)^{-(n-1)} - 2(n-1)a^2 I_n \Rightarrow 2(n-1)a^2 I_n = (2n-3)I_n + \frac{x}{(x^2 + a^2)^{n-1}}$$

34. Use Integration by parts

a) $I = \int \frac{dx}{e^{2x}+1} = \int \frac{e^{-2x}}{1+e^{-2x}} dx = -\frac{1}{2}\log(1 + e^{-2x}) + c(R)$
$$= -\frac{1}{2}\log\left(\frac{1 + e^{2x}}{e^{2x}}\right) + c = -\frac{1}{2}\{\log(1 + e^{2x}) - 2x\} + c = x - \frac{1}{2}\log(1 + e^{2x}) + c$$

b) $I = \frac{e^x dx}{\sqrt{(e^{2x}+1)}}$; Put $e^x = t \Rightarrow e^x dx = dt$

Then, $I = \int \frac{dt}{\sqrt{(t^2+1)}} = \log\{t + \sqrt{t^2+1}\} + c = \log\{e^x + \sqrt{e^{2x}+1}\} + c$(P)

$= \log\left\{e^x\left(1 + \sqrt{(1+e^{-2x})}\right)\right\} + c = x + \log\left(1 + \sqrt{(1+e^{-2x})}\right) + c$

c) $I = \int \frac{dx}{(e^{2x}+1)^2} = \int \frac{e^x dx}{e^x(e^{2x}+1)^2}$ Put $e^x = \tan\theta \therefore e^x dx = \sec^2\theta d\theta$

Then, $I = \int \frac{\sec^2\theta d\theta}{\tan\theta \cdot \sec^4\theta} = \int \frac{d\theta}{\tan\theta \cdot \sec^2\theta} = \int \frac{\cos^3\theta d\theta}{\sin\theta} = \int \frac{(1-\sin^2\theta)\cos\theta d\theta}{\sin\theta}$

Put $\sin\theta = t \Rightarrow \cos d\theta = dt$

Then, $I = \int \frac{(1-t^2)dt}{t} = \log t - \frac{t^2}{2} + c = \log\sin\theta - \frac{\sin^2\theta}{2} + c$

$= \log\left(\sqrt{\frac{e^{2x}}{e^{2x}+1}}\right) - \frac{1}{2}\frac{e^{2x}}{(e^{2x}+1)} + c = x - \frac{1}{2}\log(1+e^{2x}) + \frac{1}{2}\left(\frac{e^{2x}}{e^{2x}+1}\right) + c$(Q)

d) put $e^x = t$

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## WORKSHEET – 4

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) D | 2) A | 3) C | 4) C | 5) A | 6) A | 7) B |
| 8) B | 9) C | 10) B | | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) A | 3) A | 4) C | 5) A | 6) C | 7) C |
| 8) A | 9) C | 10) A | 11) C | 12) B | 13) B | 14) B |
| 15) B | 16) B | 17) A | 18) C | 19) B | 20) C | 21) B |
| 22) A | 23) A | 24) A,B | 25) A | 26) C | 27) A | 28) B |
| 29) *a -r*; *b-s*; *c-p*; *d-q* | | 30) C | 31) B,D | | | |

1. $\int \frac{1}{(x^2+a^2)(x^2+b^2)}dx = \int \frac{1}{b^2-a^2}\left[\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2}\right]dx$

$= \frac{1}{b^2-a^2}\int \frac{1}{x^2+a^2}dx - \frac{1}{b^2-a^2}\int \frac{1}{x^2+b^2}dx$

$= \frac{1}{b^2-a^2}\left[\frac{1}{a}\text{Tan}^{-1}\left(\frac{x}{a}\right)\right] - \frac{1}{b^2-a^2}\left[\frac{1}{b}\text{Tan}^{-1}\left(\frac{x}{b}\right)\right] + c$

2. Put $e^x = t$. Then $e^x dx = dt$.

$\int \frac{e^x}{(e^x+2)(e^x-1)}dx = \int \frac{1}{(t+2)(t-1)}dt = \int \frac{1}{3}\left[\frac{1}{t-1} - \frac{1}{t+2}\right]dt = \frac{1}{3}\int \frac{1}{t-1}dt - \frac{1}{3}\int \frac{1}{t+2}dt$

$= \frac{1}{3}\log|t-1| - \frac{1}{3}|t+2| + c = \frac{1}{3}\log\left|\frac{t-1}{t+2}\right| + c = \frac{1}{3}\log\left|\frac{e^x-1}{e^x+2}\right| + c$

3. Put $\tan x = t$. Then $\sec^2 x dx = dt \Rightarrow (1+\tan^2 x)dx = dt \Rightarrow \frac{dt}{1+t^2}\sin 2x = \frac{2t}{1+t^2}$

4. $\int \frac{1}{5+4\cos^2 x}dx = \int \frac{\sec^2 x}{5\sec^2 x+4}dx = \int \frac{\sec^2 x}{5+5\tan^2 x+4}dx = \int \frac{\sec^2 x}{9+5\tan^2 x}dx$

Put $t = \tan x$. Then $dt = \sec^2 x dx$

$$\therefore \int \frac{1}{5+4\cos^2 x}dx = \int \frac{1}{9+5t^2}dt = \int \frac{1}{3^2+(\sqrt{5}t)^2}dt$$

$$= \frac{1}{\sqrt{5}} \cdot \frac{1}{3}\text{Tan}^{-1}\left(\frac{\sqrt{5}t}{3}\right) + c = \frac{1}{3\sqrt{5}}\text{Tan}^{-1}\left(\frac{\sqrt{5}}{3}\tan x\right) + c$$

5. $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$\int \frac{dx}{a^2\sin^2 x + b^2\cos^2 x} = \int \frac{\sec^2 x dx}{a^2\tan^2 x + b^2} = \int \frac{dt}{a^2t^2+b^2} = \frac{1}{ab}\text{Tan}^{-1}\left(\frac{at}{b}\right) + c$$
$$= \frac{1}{ab}\text{Tan}^{-1}\left[\frac{a}{b}\tan x\right] + c$$

6. If $x^4 = t$, then $4x^3dx = dt \Rightarrow x^3dx = dt/4$

$$\int \frac{x^4}{\sqrt{a^8-x^8}}dt = \int \frac{dt/4}{\sqrt{a^8-t^2}} = \frac{1}{4}\int \frac{dt}{\sqrt{a^8-t^2}} = \frac{1}{4}\sin^{-1}\frac{t}{a^4} + c = \frac{1}{4}\sin^{-1}\left(\frac{x^4}{a^4}\right) + c$$

7. $\int \sqrt{\left[\frac{5-x}{x-2}\right]}dx = \int \frac{5-x}{\sqrt{(x-2)(5-x)}}dx = \int \frac{5-x}{\sqrt{-10-x^2+7x}}dx = \int \frac{\frac{1}{2}(7-2x)+\frac{3}{2}}{\sqrt{-10-x^2+7x}}$

$$= \frac{1}{2}\int \frac{7-2x}{\sqrt{-10-x^2+7x}}dx + \frac{3}{2}\int \frac{dx}{\sqrt{-10-x^2+7x}} = \frac{1}{2}\int \frac{2tdt}{\sqrt{t^2}} + \frac{3}{2}\int \frac{dx}{\sqrt{-10-(x^2-7x)}}$$

$$= \int dt + \frac{3}{2}\int \frac{dx}{\sqrt{(3/2)^2-(x-7/2)^2}} = t + \frac{3}{2}\sin^{-1}\left(\frac{x-7/2}{3/2}\right) + c$$

$$= \sqrt{7x-x^2-10} + \frac{3}{2}\sin^{-1}\left(\frac{2x-7}{3}\right) + c$$

8. $x^2 - 13x + 42 = (x-6)(x-7) = 0 \Rightarrow \int \frac{1}{x^2-13x+42}dx = \int \frac{1}{x-7}dx - \int \frac{1}{x-6}dx$

9. $I = \int \frac{e^x}{e^{2x}+2-3e^x}dx \Rightarrow$ Let $= e^x dx = dt$

$$\Rightarrow I = \int \frac{dt}{t^2-3t+2}$$

$$= \int \frac{dt}{(t-1)(t-2)} = \int \frac{1}{(t-2)}dt - \int \frac{1}{(t-1)}dt = \log\left|\frac{t-2}{t-1}\right| + c = \log\left|\frac{e^x-2}{e^x-1}\right| + c$$

10. Put $e^x - 1 = t$. Then $e^x dx = dt$.

$$\int \frac{1}{(e^x-1)^2}dx = \int \frac{e^x}{e^x(e^x-1)^2}dx = \int \frac{1}{(t+1)t^2}dt = \int -\frac{1}{t} + \frac{1}{t^2} + \frac{1}{t+1}$$

$$= -\log|t| - \frac{1}{t} + \log|t+1| = c = -\frac{1}{e^x-1} + \log\left|\frac{e^x}{e^x-1}\right| + c$$

11. $\int \sqrt{2x^2+3x+4}dx = \sqrt{2}\int \sqrt{x^2+\frac{3}{2}x+2}dx = \sqrt{2}\int \sqrt{\left(x+\frac{3}{4}\right)^2+2-\frac{9}{16}}dx$

$$= \sqrt{2}\int \sqrt{\left(x+\frac{3}{4}\right)^2+\left(\frac{\sqrt{23}}{4}\right)}dx$$

12. $\int \frac{1}{\sqrt{4x-3-x^2}}dx = \int \frac{1}{\sqrt{1^2-(x-2)^2}}dx$ put $x-2=t$

13. $\int \frac{\cos x+x\sin x}{x(x-\cos x)}dx = \int \frac{x(1+\sin x)+(\cos x-x)}{x(x-\cos x)}dx = \int \frac{1+\sin x}{x-\cos x}dx - \int \frac{x-\cos x}{x(x-\cos x)}dx$

$= \log|x-\cos x| - \log|x| + c = \log\left|\frac{x-\cos x}{x}\right| + c = \log\left|1-\frac{\cos x}{x}\right| + c$

14. $9\cos x - \sin x = A\frac{d}{dx}[4\sin x + 5\cos x] + B[4\sin x + 5\cos x]$

$\Rightarrow 9\cos x - \sin x = A[4\cos x - 5\sin x] + B[4\sin x + 5\cos x]$

$4\,A + 5\,B = 9$

$-5A + 4B = -1$

Solve (i) & (ii)

$20\,A + 25\,B = 45$

$-20\,A + 16\,B = -4$

$\Rightarrow 41\,B = 41$ $\Rightarrow B = 1, A = 1.$

15. Put $x = \frac{1}{t}$, then $dx = -\frac{1}{t^2}dt$

$$\int \frac{1}{(x^2+9)\sqrt{x^2-9}}dx = \int \frac{1}{\left(\frac{1}{t^2}+9\right)\sqrt{\frac{1}{t^2}-9}}\left(-\frac{1}{t^2}\right)dt = \int \frac{-t^2\cdot 1}{(x+9t^2)\sqrt{1-9t^2}}\left(\frac{1}{t^2}\right)dt$$

16. $5\sin x = A(\sin x - 2\cos x) + B\frac{d}{dx}(\sin x - 2\cos x)$

$2x+3 = 2[3-t^2]+3 = 6-2t^2+3 = 9-2t^2$

$$\int \frac{2x+3}{\sqrt{3-x}}dx = \int \frac{9-2t^2}{t}\cdot(-2t)dt = \int (4t^2-18)dt = 4\int t^2dt - 18\int 1dt$$

$$= \frac{4t^3}{3} - 18t + c = \frac{4}{3}(3-x)^{3/2} - 18\sqrt{3-x} + c$$

17. Put $\cos x = t$

18. Put $x^9 - 1 = t$

19. put $\tan\frac{x}{2} = t$ $dx = \frac{2dt}{1+t^2}$, $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$

$$\therefore \int \frac{1}{1+\sin x+\cos x}dx = \int \frac{\frac{2dt}{1+t^2}}{1+\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}} = 2\int \frac{1}{1+t^2+2t+1}dt$$

$$= 2\int \frac{1}{2t+1}dt = \int \frac{1}{1+t}dt = \log|1+t| + C$$

20. $2x+5 = A\frac{d}{dx}(x^2-2x+10) + B$

21. Resolve into Partial Fractions

22. $I = \int \frac{1}{x(x^{10}-1)}dx$ 23. $\frac{1}{3}\int \frac{1}{x^2+1}dx + \frac{2}{3}\int \frac{1}{x^2+4}dx$

23. $\int \frac{1}{x-x^3}dx = \int \frac{1}{x(1-x)(1+x)}dx = \int \frac{1}{x}dx + \frac{1}{2}\int \frac{1}{1-x}dx$

$\Rightarrow \log x - \frac{1}{2}\log(1-x) - \frac{1}{2}\log(1+x) \Rightarrow \frac{1}{2}[\log x^2 - \log(1-x) - \log(1+x)]$

$\Rightarrow \frac{1}{2}\log\left|\frac{x^2}{1-x^2}\right| + c$

25. Conceptual.

26. $3\cos x + 2\sin x = A(4\cos x - 5\sin x) + B(4\sin x + 5\cos x)$

$\Rightarrow 4A + 5B = 3; -5A + 4B = 2 \Rightarrow A = \frac{2}{41}$ and $B = \frac{23}{41}$

$\therefore \int \frac{3\cos x + 2\sin x}{4\sin x + 5\cos x}dx = \frac{2}{41}\int \frac{4\cos x - 5\sin x}{4\sin x + 5\cos x}dx + \frac{23}{41}\int \frac{5\cos x}{dx}$

$\Rightarrow \frac{2}{41}\log|4\sin x + 5\cos x| + \frac{23}{41}x + c$

27. $\sin x + 8\cos x = A(4\sin x + 6\cos x) + B(4\cos x - 6\sin x) \Rightarrow A = 1, B = \frac{1}{2}$

$\therefore \int \frac{\sin x + 8\cos x}{4\sin x + 6\cos x}dx = \int 1.dx + \frac{1}{2}\int \frac{4\cos x - 6\sin x}{4\sin x + 6\cos x}dx = x + \frac{1}{2}\log|4\sin x + 6\cos x| + c.$

28. $3\cos x - 2\sin x = A[4\cos x - 5\sin x] + B(4\sin x + 5\cos x)$

$\Rightarrow A = \frac{22}{41}; B = \frac{7}{41}$

$\therefore \int \frac{3\cos x - 2\sin x}{4\sin x + 5\cos x} = \frac{22}{41}\int \frac{4\cos x - 5\sin x}{4\sin x + 5\cos x} + \frac{7}{41}\int 1dx$

$= \frac{22}{41}\log|4\sin x + 5\cos x| + \frac{7}{41}x + c$

29. a) $\int \frac{1}{1+\cos x}dx = \int \frac{1-\cos x}{\sin^2 x}dx = \int \text{cosec}^2 x dx - \int \cot x \text{cosec}\, x dx = -\cot x + \text{cosec}\, x$

b) $\int \frac{1}{1-\cos x}dx = \int \frac{1+\cos x}{\sin^2 x}dx = \int \text{cosec}^{\ 2} x dx - \int \text{cosec}\, x \cot dx = -\cot x + \text{cosec}\, x + C$

c) $\int \frac{1}{1-\sin x}dx = \int \frac{1+\sin x}{\cos^2 x}dx = \int \sec^2 x dx + \int \sec^2 x dx \int \text{Tan}\, x \sec x dx = \text{Tan}\, x + \sec x + C$

d) $\int \frac{1}{1+\sin x}dx = \int \frac{1-\sin x}{\cos^2 x}dx = \int \sec^2 x dx - \int \text{Tan}\, x \sec x dx = \text{Tan}\, x - \sec x + C$

30. $dx = \frac{2dt}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2} \ \therefore \int \frac{1}{4+5\left(\frac{1-t^2}{1+t^2}\right)} \times \frac{2dt}{1+t^2} = \int \frac{2dt}{4+4t^2+5-5t^2} = 2\int \frac{1}{9-t^2}dt$

$\Rightarrow 2\left[\frac{1}{2\times 3}\log\left|\frac{3+t}{3-t}\right| + C\right] = \frac{1}{3}\log\left|\frac{3+t}{3-t}\right| + C$

31. $\int \frac{dx}{x^4+x^3} = \int \frac{(x+1)}{x^3(x+1)}dx = \int \left[\frac{1}{x^3} - \frac{1}{x^2(x-1)}\right]dx$

$= \int \left[\frac{1}{x^3} - \frac{1}{x^2} + \frac{1}{x(x+1)}\right]dx = \int \left[\frac{1}{x^3} - \frac{1}{x^2} + \frac{1}{x} + \frac{1}{x+1}\right]dx$

$$= -\frac{1}{2x^2} + \frac{1}{x} + \log x|x| - \log|x+1| + c = \frac{1}{2x^2} + \frac{1}{x} + \left|\frac{x}{x+1} + c\right|$$

32.

a) $\int \frac{1}{\sqrt{x^2-4x-5}}dx = \int \frac{1}{\sqrt{(x-2)^2-(3)^2}}dx = \cosh^{-1}\left(\frac{x-2}{3}\right) + c$

b) $\int \frac{1}{\sqrt{-x^2+2x+3}}dx = \int \frac{1}{\sqrt{(2)^2-(x-1)^2}}dx = \sin^{-1}\left(\frac{x-1}{2}\right) + c$

c) $\int \frac{1}{\sqrt{9x-6-x^2}}dx = \int \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2-\left(x-\frac{5}{2}\right)^2}}dx = \sin^{-1}(2x-5) + c$

d) $\int \frac{1}{\sqrt{9x^2-25}}dx = \int \frac{1}{\sqrt{(3x)^2-(5)^2}}dx = \frac{1}{3}\cosh^{-1}\left(\frac{3x}{5}\right) + c$

33. a) $\mathrm{dx} = \frac{2\mathrm{dt}}{1+\mathrm{t}^2}, \cos \mathrm{x} = \frac{1-\mathrm{t}^2}{1+\mathrm{t}^2} \therefore \int \frac{1}{4+5\left(\frac{1-\mathrm{t}^2}{1+\mathrm{t}^2}\right)} \times \frac{2\mathrm{dt}}{1+\mathrm{t}^2} = \int \frac{2\mathrm{dt}}{4+4\mathrm{t}^2+5-5\mathrm{t}^2} = 2\int \frac{1}{9-\mathrm{t}^2}\mathrm{dt} \Rightarrow$

$2\left[\frac{1}{2\times 3}\log\left|\frac{3+\mathrm{t}}{3-\mathrm{t}}\right| + \mathrm{C}\right] = \frac{1}{3}\log\left|\frac{3+\mathrm{t}}{3-\mathrm{t}}\right| + \mathrm{C}$

b) $\int \frac{1}{5+4\left(\frac{2\mathrm{t}}{1+\mathrm{t}^2}\right)} \times \frac{2\mathrm{dt}}{1+\mathrm{t}^2} = 2\int \frac{1}{5+5\mathrm{t}^2+8\mathrm{t}}\mathrm{dt}$

$$\Rightarrow \int \frac{1}{\left(t+\frac{4}{5}\right)^2 + \left(\frac{\sqrt{11}}{5}\right)^2}dt = \frac{2}{5}\left[\frac{5}{\sqrt{3}}\mathrm{Tan}^{-1}\left(\frac{5t+4}{\sqrt{3}}\right)\right] + C = \frac{2}{\sqrt{3}}\mathrm{Tan}^{-1}\left[\frac{5\tan\frac{x}{2}+4}{\sqrt{3}}\right] + C$$

c) $3\cos x + 2\sin x = A\frac{d}{dx}[4\cos x + 3\sin x] + B[4\cos x + 3\sin x]$

$\Rightarrow 3\cos x + 2\sin x = A[-4\sin x + 3\cos x] + B[4\cos x + 3\sin x]$

$$\Rightarrow -4A + 3B = 2$$
$$3A + 4B = 3$$

Solving (i) & (ii)

$$-12A + 9B = 6$$
$$12\,\mathrm{A} + 16\,\mathrm{B} = 12$$
$$B = \frac{18}{25}$$
$$\Rightarrow -4\,\mathrm{A} + \frac{54}{25} = 2 \Rightarrow -4\,\mathrm{A} = 2 - \frac{54}{25}$$
$$\Rightarrow -4\,\mathrm{A} = \frac{-4}{25} \Rightarrow \mathrm{A} = \frac{1}{25}$$
$$\int \frac{3\cos x + 2\sin x}{4\cos x + 3\sin x}dx = \frac{1}{25}\log|4\cos x + 3\sin x| + \frac{18}{25}x + C$$

d) $3\cos x - 2\sin x = A[4\cos x - 5\sin x] + B[4\sin x + 5\cos x]$

$4A + 5B = 3$

Solve (i) & (ii)

$$20\,\mathrm{A} + 25\,\mathrm{B} = 15$$
$$-20\,\mathrm{A} + 16\,\mathrm{B} = -8$$
$$41\,\mathrm{B} = 7$$

$B = \frac{7}{41}$ and $A = \frac{22}{41}$.

34. $I = \int \frac{\sin x}{\sin 4x} dx = \frac{1}{4}\int \frac{\cos x}{\cos^2 x \cos 2x} dx = \frac{1}{4}\int \frac{\cos x}{(1-\sin^2 x)(1-2\sin^2 x)} dx$

$$= \frac{1}{4}\frac{dt}{(1-t^2)(1-2t^2)} [\text{ where } t = \sin x]$$

$$= \frac{1}{4}\int \left[\frac{2}{1-2t^2} - \frac{1}{1-t^2}\right] dt = \frac{1}{2}\cdot\frac{1}{2\sqrt{2}}\log\left|\frac{1+\sqrt{2}t}{1-\sqrt{2}t}\right| - \frac{1}{8}\log\left|\frac{1+t}{1-t}\right| + c$$

$$= \frac{1}{4\sqrt{2}}\log\left|\frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x}\right| - \frac{1}{8}\log\left|\frac{1+\sin x}{1-\sin x}\right| + c \Rightarrow A = -\frac{1}{8}, B = \frac{1}{4\sqrt{2}}\Bigg]$$

# 5. Definite Integrals

## WORKSHEET – 1

### Level – I

1) B 2) B 3) D 4) A 5) B 6) B

7) B 8) C 9) B 10) C

1. $\left[3\frac{x^3}{3}+4\frac{x^2}{2}+3x\right]_0^2 = 22-0 = 22.$ 2. $[\tan x]_0^{\frac{\pi}{4}} = 1-0 = 1.$

3. $[\sin x - \cos x]_0^{\frac{\pi}{2}} = (1-0)-(0-1) = 2.$ 4. $[\log(1+x)]_0^1 = \log 2 - \log 1 = \log 2.$

4. $\frac{1}{2}\int_0^1 \frac{2x}{1+x^2}\cdot dx = \frac{1}{2}[\log(1+x^2)]_0^1 = \frac{1}{2}[\log 2 - \log 1] = \frac{1}{2}\log 2.$

5. $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \sin x dx = 0$ as, sin x is an odd function.

6. $\int_0^1 \frac{4x^3}{\sqrt{1-(x^4)^2}}dx$ put $x^4 = t \Rightarrow 4x^3dx = dt \Rightarrow x = 0 \Rightarrow t = 0 \Rightarrow x = 1 \Rightarrow t = 1$

$\int_0^1 \frac{1}{\sqrt{1-t^2}}dt = [\sin^{-1} t]_0^1 = \sin^{-1} 1 - \sin^{-1} 0 = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$

8. $\int_0^{\frac{\pi}{4}} \sec^2 x \tan^2 x dx = \left[\frac{\tan^3 x}{3}\right]_0^{\frac{\pi}{4}} = \frac{1}{3}.$

9. $\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1} + \cot^{-1}\frac{x}{x^2+1}\right)dx = \int_{-1}^{3}\frac{\pi}{2}dx = \frac{\pi}{2}(x)_{-1}^3 = \frac{\pi}{2}[(3-(-1))] = 2\pi.$

10. $\log x = t \Rightarrow \frac{1}{x}dt = dt \Rightarrow x = 1 \Rightarrow t = 0 \Rightarrow x = 2 \Rightarrow t = \log 2$

$$\int_1^2 \frac{\log x}{x}\cdot dx = \int_0^{\log 2} t\cdot dt = \left(\frac{t^2}{2}\right)_0^{\log 2} = \frac{(\log 2)^2}{2}$$

### Level – II

1) 2 2) 1 3) 3 4) 3 5) 3 6) 19

7) 4 8) 3 9) 1 10) 4 11) 1 12) 3

13) 2 14) 4 15) 4 16) 1 17) 3 18) 4

19) 3,4 20) 1 21) 2 22) 3 23) 3

24) a-q; b-r; c-p; d-s 25) 1,2,3 26) 0 27) a-s; b-s; c-p; d-r

28) 1,2,4 29) 2 30) a-r; b-t; c-q; d-r

1. Let $\log x = t \Rightarrow \frac{dx}{x} = dt$ if $x = 1, t = 0$ & if $x = e, t = 1 \Rightarrow \int_1^e \frac{(\log x)^3}{x}dx = \int_0^1 t^3 dt = \left[\frac{t^4}{4}\right]_0^1 = \frac{1}{4}.$

2. Let, $\log x = t \Rightarrow \frac{1}{x}dx = dt \Rightarrow x = 1 \Rightarrow t = 0 \Rightarrow x = 2 \Rightarrow t = \log 2$

$\int_1^2 \frac{\cos(\log x)}{x}dx = \int_0^{\log 2}\cos t\cdot dt = (\sin t)_0^{\log 2} = \sin(\log 2).$

3. $\int_0^{\frac{\pi}{4}} \frac{\sin^9 x}{\cos^{11} x} dx = \int_0^{\frac{\pi}{4}} \tan^9 x \sec^2 x dx = \left[\frac{\tan^{10} x}{10}\right]_0^{\frac{\pi}{4}} = \frac{1}{10}.$

4. $I = \int_0^a \frac{f(x)}{f(x)+f(a-x)} dx = \int_0^a \frac{f(a-x)}{f(a-x)+f(x)} dx \Rightarrow 2I = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}.$

5. $\int_0^{\pi/2} \frac{(\sin x+\cos x)^2}{\sqrt{1+\sin x}} dx = \int_0^{\pi/2} \frac{(\sin x+\cos x)^2}{\sqrt{1+\sin x}} dx = \int_0^{\pi/2} (\sin x + \cos x) dx = (-\cos x + \sin x)_0^{\pi/2}$

$= [-0 + 1] - [-1 + 0] = 2.$

6. $\int_{-2}^{-1} f(x)dx + \int_{-1}^{0} f(x)dx + \int_0^1 f(x)dx + \int_1^2 f(x)dx + \int_2^3 f(x)dx + \int_3^4 f(x)dx$

$= (-2)^2 + (-1)^2 + (0)^2 + 1^2 + 2^2 + 3^2 = 4 + 1 + 0 + 1 + 4 + 9 = 19.$

7. let $3 + 4\sin x = t \Rightarrow 4\cos x \cdot dx = dt \Rightarrow$ if $x = 0$ then $t = 3 \Rightarrow x = \frac{\pi}{3}$ then $t = 3 + 2\sqrt{3}$

$$I = \frac{1}{4}\int_3^{3+2\sqrt{3}} \frac{1}{t} dt = \frac{1}{4}[\log t]_3^{3+2\sqrt{3}} = \frac{1}{4}\log\left(\frac{3+2\sqrt{3}}{3}\right).$$

8. let $I = \int_0^{\frac{\pi}{2}} \frac{\sin^5 x}{\sin^5 x+\cos^5 x} \cdot dx \Rightarrow f(x) = f(a-x)$

$$I = \int_0^{\pi_2} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} \cdot dx \Rightarrow 2I = \int_0^{\pi_2} 1.dx \Rightarrow 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}.$$

9. $\int_0^1 x \cdot \tan^{-1} \cdot dx = \left(\frac{x^2}{2}\tan^{-1} x\right)_0^1 - \int_0^1 \frac{x^2}{2}\frac{1}{1+x^2} \cdot dx \Rightarrow \frac{1}{2} \cdot \frac{\pi}{4} - \frac{1}{2}\left[\int_0^1 \left(\frac{x^2+1}{x^2+1} - \frac{1}{x^2+1}\right) \cdot dx\right]$

$$\frac{\pi}{8} - \frac{1}{2}[[x]_0^1 - [\tan^{-1} x]_0^1] \Rightarrow \frac{\pi}{8} - \frac{1}{2}\left[1 - \frac{1}{2} \cdot \frac{\pi}{2}\right] = \frac{\pi}{8} + \frac{\pi}{8} - \frac{1}{2} = \frac{\pi}{4} - \frac{1}{2}$$

10. $\int_0^{\frac{\pi}{4}} \tan^2 x(1 + \tan^2 x)dx = \int_0^{\frac{\pi}{4}} \tan^2 x \sec^2 x dx = \left[\frac{\tan^3 x}{x}\right]_0^{\frac{\pi}{4}} = \frac{1}{3} \Rightarrow \frac{a}{b} = \frac{1}{3} \Rightarrow \text{a} + \text{b} = 4.$

11. $\int_0^1 x(1-x)^n dx = \int_0^1 (1-x)x^n \cdot dx = \int_0^1 (x^n - x^{n+1}) \cdot dx = \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}\right]_0^1 = \frac{1}{n+1} - \frac{1}{n+2}.$

12. Let $\sin \text{x} = \text{t} \Rightarrow \cos \text{x} \text{dx} = \text{dt} \int_0^{\text{k}} \frac{\cos \text{x}}{1+\sin^2 \text{x}} \text{dx} = \int_0^{\text{k}} \frac{\text{dt}}{1+\text{t}^2} = [\tan^{-1} \text{t}]_0^{\sin \text{k}} = \tan^{-1}(\sin \text{k})$
$\tan^{-1}(\sin k) = \frac{\pi}{4} \Rightarrow \sin k = 1 \Rightarrow k = \frac{\pi}{2}.$

13. $\int_0^{\pi} \frac{x}{1+\sin x} dx = \frac{\pi}{2}\int_0^{\pi} \frac{1}{1+\sin x} dx = \frac{\pi}{2} \times 2\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x} dx = \pi\int_0^{\frac{\pi}{2}} \frac{1-\sin x}{\cos^2 x} dx = \pi\int_0^{\frac{\pi}{2}} (\sec^2 x - \sec x \cdot \tan x)dx = \pi[\tan x - \sec x]_0^{\frac{\pi}{2}} = \pi[0 - (0 - 1)] = \pi.$

14. let $I = \int_0^{\frac{\pi}{2}} \sin^{100} x - \cos^{100} x \cdot dx \Rightarrow I = \int_0^{\frac{\pi}{2}} \left[\sin^{100}\left(\frac{\pi}{2} - x\right) - \cos^{100}\left(\frac{\pi}{2} - x\right)\right] dx$

$$I = \int_0^{\frac{\pi}{2}} (\cos^{100} x - \sin^{100} x)dx \Rightarrow 2I = 0 \Rightarrow I = 0$$

15. $I = \int_0^{\frac{\pi}{4}} \frac{\sin x+\cos x}{7+9\sin 2x} \cdot dx \Rightarrow$ sub $\cos x - \sin x = t$ then $-(\cos x + \sin x)dx = dt.$

If $x = 0$, then $t = 1$. If $x = \frac{\pi}{4}$, then $t = 0 \Rightarrow I = \int_0^1 \frac{-1}{16-9t^2} \cdot dt = \left[\frac{1}{24}\log\left|\frac{4+3t}{4-3t}\right|\right]_0^1 = \frac{1}{24}\log 7.$

16. $I = \frac{\pi}{2}\int_0^{\pi} \frac{1}{a\cos^2 x + b\sin^2 x}dx = \frac{\pi}{2}\frac{1}{ab}\tan^{-1}\left(\frac{b}{a}\tan x\right)_0^{\pi} = \frac{\pi}{2ab}(\pi - 0) = \frac{\pi^2}{2ab}.$

17. $\int_0^{in5} \frac{e^x\sqrt{e^x-1}}{e^x+3}\cdot dx$ let $e^x - 1 = t^2$ $x = 0 \Rightarrow t = 0 \Rightarrow x = \ln 5 \Rightarrow t = 2$

$$e^x \cdot dx = 2tdt \Rightarrow \int_0^2 \frac{t}{t^2+4}2tdt = 2\int_0^2 \frac{t^2}{t^2+4}\cdot dt$$

$= 2\left[\int_0^2 \left(1 - \frac{4}{t^2+4}\right)dt\right] = 2\left[[t]_0^2 - 2\left[\tan^{-1}\frac{t}{2}\right]_0^2\right] = 2\left[2 - 2\left(\frac{\pi}{4}\right)\right] = 4 - \pi.$

18. $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{3 + \sin 2x}dx = \int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{4 - (\sin x - \cos x)^2}$ put $\sin x - \cos x = t$

$(\cos x + \sin x)\cdot dx = dt \Rightarrow$ If $x = 0, \frac{\pi}{4}$ then $t = -1, 0 \Rightarrow I = \int_{-1}^{0} \frac{dt}{4-t^2} = \left[\frac{1}{4}\log\frac{2+t}{2-t}\right]_{-1}^{0} = \frac{1}{4}\log 3.$

19. $\int_0^{\frac{\pi}{4}} \tan^2 x \cdot dx = \int_0^{\frac{\pi}{4}} (\sec^2 x - 1)dx = [\tan x - x]^{\frac{\pi}{4}} = 1 - \frac{\pi}{4}.$

20. Stmt 1: $-= \int_0^{2\pi} \cos^{99} x \cdot dx = 2\int_0^{\pi} \cos^{99} x = 2x\int_0^{2\pi} \cos^{99}(\pi - x)dx$

$= 2 \times 0[\because \cos^{99}(\pi - x) = -\cos^{99} x] = 0.$ Stmt 2: $-$ true.

21. Let $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}$ $\left[\because \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$

Now , $2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1dx = [x]^{\frac{\pi}{3}}\frac{\pi}{6} \Rightarrow I = \frac{\pi}{6}.$

22. $\int_{-a}^{a} f(x)dx = 0$ if $f(-x) = -f(x)$ i.e $f(x)$ is an odd function.

$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x^3 dx = 0$ as $\sin x^3$ is an odd function.

23. $I = \int_0^{2\pi} \sin^{100} x\cos^{99} x dx$ here $f(x) = \sin^{100} x\cos^{99} x$ for which $f(2\pi - x) = f(x)$

or $I = 2\int_0^{\pi} \sin^{100} x\cos^{99} x dx = 2\int_0^{\pi} \sin^{100}(\pi - x)\cos^{99}(\pi - x)dx$

$= -2\int_0^{\pi} \sin^{100} x\cos^{99} x dx =$ I (or) 2I = 0, or I = 0.

24. (a) $\frac{1}{\pi}\left[\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)\right]_{-2}^{2} = \frac{1}{2\pi}\left[\frac{\pi}{4} + \frac{\pi}{4}\right] = \frac{1}{4}.$

(b) $[\sec^{-1} x]_{\sqrt{2}}^{2} = \sec^{-1} 2 - \sec^{-1}\sqrt{2} = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}.$

(c) $\frac{1}{2}[\tan^{-1} 2x]_0^k = \frac{\tan^{-1} 2k}{2} \times \frac{1}{2} = \frac{\pi}{16}$ $\therefore \tan^{-1} 2k = \frac{\pi}{4} \Rightarrow k = \frac{1}{2}.$

(d) $\frac{1}{2}[\log(2x+1)]_0^{40} = \frac{1}{2}[\log 81] = \log 9.$

25. $\int_0^{\pi/4} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x} = \int_0^{\pi/4} \frac{\sec^2 x}{a^2 + b^2\tan^2 x}dx$

Put tan x = t then $\sec^2$ xdx = dt

If x = $0, \pi/4$ then t = 0,1

$= \int_0^1 \frac{1}{a^2+b^2t^2} \cdot dt = \frac{1}{b^2}\int_0^1 \frac{1}{t^2+\left(\frac{a}{b}\right)^2} \cdot dt = \frac{1}{b^2} \cdot \frac{1}{\frac{a}{b}} \cdot \left[\tan^{-1}\left(\frac{bt}{a}\right)\right]_0^1 = \frac{1}{ab}\tan^{-1}\left(\frac{b}{a}\right).$

26. $\log\frac{2+x}{2-x}$ is an odd fuction $\therefore \int_{-1}^{1} \log\frac{2+x}{2-x} dx = 0.$

27. $a \to s, b \to s, c \to p, d \to r$

a) $\int_{-1}^{1} \frac{1}{1+x^2} dx = (\tan^{-1} x)_{-1}^{1} = \tan^{-1} 1 - \tan^{-1}(-1) = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}.$

b) $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} dx = (\sin^{-1} x)_0^1 = \sin^{-1}(1) - \sin^{-1}(0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$

c) $\int_2^3 \frac{dx}{1-x^2} = \left[\frac{1}{2}\log\left|\frac{1+x}{1-x}\right|\right]_2^3 = \frac{1}{2}(\log 2 - \log 3) = \frac{1}{2}\log\left(\frac{2}{3}\right).$

d) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}} = [\sec^{-1} x]_1^2 = \sec^{-1} 2 - \sec^{-1} 1 = \frac{\pi}{3} - 0 = \frac{\pi}{3}.$

28. $I = \int_{\tan^{-1}\lambda}^{\cot^{-1}\lambda} \frac{\tan x}{\tan x+\cot x} dx = \int_{\tan^{-1}\lambda}^{\cot^{-1}\lambda} \frac{\cot x}{\tan x+\cot x} dx \Rightarrow 2I = \cot^{-1}\lambda - \tan^{-1}\lambda$

$2I = \frac{\pi}{2} - 2\tan^{-1}\lambda \Rightarrow \text{Tan}^{-1}\lambda = \frac{\pi}{4} - I \therefore I \in \left(\frac{-\pi}{4}, \frac{3\pi}{4}\right)$, then I can't take the value $\frac{-\pi}{4}, \frac{-\pi}{2}, \frac{3\pi}{4}$.

29. $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}} dx = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx \Rightarrow 2I = \int_2^3 (1) dx = (x)_2^3 = 1 \Rightarrow I = \frac{1}{2}.$

30. Let $I_1 = \int_0^1 \frac{e^{\tan^{-1}x}}{\sqrt{1+x^2}} \cdot dx \Rightarrow$ Put $\tan^{-1} x = t \Rightarrow \frac{1}{1+x^2} \cdot dx = dt \Rightarrow \frac{dx}{\sqrt{1+x^2}} = \sqrt{1+x^2} \cdot dt$

$\frac{dx}{\sqrt{1+x^2}} = \sec t. dt$

a) Now $I_1 = \int_0^{\pi/4} e^t \cdot \sec t \cdot dt \; I_2 = \int_0^{\pi/4} e^t \cdot \sec t \cdot \tan t \cdot dt$

$I_1 = (e^t \cdot \sec t)_0^{\pi/4} - I_2 \;\; \therefore I_1 + I_2 = \sqrt{2} \cdot e^{\pi/4} - 1.$

b) $I_3 = \int_0^{\pi/4} e^\theta \cdot \tan^2\theta \cdot \sec\theta \cdot d\theta = \int_0^{\pi/4} e^\theta \tan\theta \cdot d(\sec\theta)$

$= \left[e^\theta \cdot \tan\theta \cdot \sec\theta\right]_0^{\pi/4} - I_2 - \int_0^{\pi/4} e^\theta \cdot \sec^3\theta d\theta = \sqrt{2}e^{\pi/4} - I_3 - I_1 - I_2$

$\therefore I_1 + I_2 + 2I_3 = \sqrt{2}e^{\pi/4}. \;\; \because I_1 + I_2 = \sqrt{2} \cdot e^{\pi/4} - 1 \Rightarrow I_3 = \frac{1}{2}$

$\therefore I_1 + I_2 - 4I_3 = \sqrt{2}e^{\pi/4} - 3.$

c) $\frac{I_1+I_2+1}{I_1+I_2+2I_3} = 1$

d) $\frac{2(I_1+I_2)}{2I_3+1} = \frac{2\left(\sqrt{2}e^{\frac{\pi}{4}}-1\right)}{2} = \sqrt{2}e^{\frac{\pi}{4}} - 1.$

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## WORKSHEET – 2

### Level – I

| | | | | | |
|---|---|---|---|---|---|
| 1) C | 2) A | 3) C | 4) B | 5) C | 6) D |
| 7) A | 8) D | 9) D | 10) A | | |

1. $\int_{0.5}^{4.5}[x]dx = \int_{0.5}^{1}[x]dx + \int_{1}^{2}[x]dx + \int_{2}^{3}[x]dx + \int_{3}^{4}[x]dx + \int_{4}^{4.5}[x]dx$

$= 0 + \int_{1}^{2}1dx + \int_{2}^{3}2dx + \int_{3}^{4}3dx + \int_{4}^{4.5}4dx = 0 + [x]_1^2 + [2x]_2^3 + [3x]_3^4 + [4x]_4^{4.5}$

$= (2-1) + (6-4) + (12-9) + (18-16) = 8.$

2. $\int_0^{50}(x-[x])dx = 50\int_0^1(x-[x])dx (\because x-[x]$ is a periodic with period 1)

$= 50\int_0^1 xdx = 50\left(\frac{x^2}{2}\right)_0^1 = 50\left(\frac{1}{2}\right) = 25.$

3. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin|x|dx = \int_{-\frac{\pi}{2}}^{0}\sin(-x)dx + \int_0^{\frac{\pi}{2}}\sin x dx = [\cos x]_{-\frac{\pi}{2}}^{0} + [-\cos x]_0^{+\frac{\pi}{2}} \Rightarrow 1 - 0 + (-0) + 1 = 2.$

4. $\int_0^{\frac{\pi}{2}}\sin^9 x dx = \frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot 1 = \frac{128}{315}.$ 5. $\int_0^{\frac{\pi}{2}}\cos^7 x dx = \frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot 1 = \frac{16}{35}.$

5. $\int_0^{\frac{\pi}{2}}\sin^8 x\cdot\cos^2 x dx = \frac{1}{10}\times\frac{7}{8}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2} = \frac{7\pi}{512}.$

6. $\int_0^{\frac{\pi}{4}}\tan^5 x dx = \frac{1}{4} - \frac{1}{2} + \frac{1}{2}\log 2 = \frac{1}{2}\log 2 - \frac{1}{4}.$

7. $\int_0^{\pi}[\cos x]dx = \int_0^{\frac{\pi}{2}}[\cos x]dx + \int_{\frac{\pi}{2}}^{\pi}[\cos x]dx = \int_0^{\pi}0\cdot dx + \int_{\frac{\pi}{2}}^{\pi}(-1)dx = 0 + (-x)_{\frac{\pi}{2}}^{\pi} = \frac{-\pi}{2}.$

8. $Lt_{n\to\infty}\sum_{r=0}^{n-1}\frac{n}{n^2+r^2} = Lt_{n\to\infty}\frac{1}{n}\cdot\sum_{r=0}^{n-1}\frac{1}{1+\left(\frac{r}{n}\right)^2} = \int_0^1\frac{1}{1+x^2}dx = [\text{Tan}^{-1}x]_0^1 = \frac{\pi}{4}$

9. $Lt_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n+r} = Lt_{n\to\infty}^{\frac{1}{n}}\cdot\sum_{r=0}^{n-1}\frac{1}{1+\left(\frac{r}{n}\right)} = \int_0^1\frac{1}{1+x}dx = [\log(1+x)]_0^1 = \log 2.$

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) A | 3) B | 4) A | 5) C | 6) A | 7) 70 |
| 8) B | 9) B | 10) B | 11) D | 12) D | 13) A | 14) B |
| 15) D | 16) B | 17) A | 18) D | 19) 6 | 20) 0 | 21) B |
| 22) B,C | 23) A | 24) B | 25) B | 26) C | 27) a-r; b-p; c-q; d-s | |
| 28) B,C | 29) a-s; b-q; c-r; d-p | | 30) A,D | 31) D | 32) a-r; b-s; c-q; d-q | |

1. $\int_0^{100\pi}\sqrt{1-\cos 2x\cdot dx} = \int_0^{100\pi}\sqrt{2\sin^2 x\cdot dx} = \sqrt{2}\int_0^{100\pi}|\sin x|\cdot dx = 100\sqrt{2}\int_0^{\pi}|\sin x|dx$

$= 100\sqrt{2}\int_0^{\pi}\sin x dx = 100\sqrt{2}(-\cos x)_0^{\pi} = 100\sqrt{2}(1+1) = 200\sqrt{2}.$

2. $\int_0^{\frac{\pi}{2}}\tan^5 x\cdot\cos^8 x dx = \int_0^{\frac{\pi}{2}}\sin^5 x\cdot\cos^3 x\cdot dx = \frac{4\times2\times2}{8\times6\times4\times2} = \frac{1}{24}.$

3. $\int_1^4\ln[x]dx = \int_1^2\ln[x]dx + \int_2^3\ln[x]dx + \int_3^4\ln[x]dx = \int_1^2 0\cdot dx + \int_2^3\ln 2dx + \int_3^4\ln 3dx = 0 + \ln 2 + \ln 3 = \ln 6.$

4. $\int_0^2[x^2-1]dx = \int_0^1[x^2-1]dx + \int_1^{\sqrt{2}}[x^2-1]dx + \int_{\sqrt{2}}^{\sqrt{3}}[x^2-1]dx + \int_{\sqrt{3}}^2[x^2-1]dx$

$= \int_0^1 (-1)dx + \int_1^{\sqrt{2}} (0)dx + \int_{\sqrt{2}}^{\sqrt{3}} (1)dx + \int_{\sqrt{3}}^{2} (2)dx = -1(1-0) + 0 + 1(\sqrt{3} - \sqrt{2}) + 2(2 - \sqrt{3}) = 3 - \sqrt{3} - \sqrt{2}.$

5. $\sum_{n=1}^{100} \int_{n-1}^{n} e^{x-[x]} \cdot dx = \int_0^1 e^x dx + \int_1^2 e^{x-1} dx + \int_2^3 e^{x-2} dx + \cdots \ldots \ldots + \int_{99}^{100} e^{x-99} dx = (e^x)_0^1 + (e^{x-2})_1^2 + \cdots \ldots \ldots + (e^{x-99})_{99}^{100} = (e-1) + (e-1) + \cdots \ldots \ldots + (e-1). = 100(e-1)$

6. $\int_0^{\pi} \sin^8 \frac{x}{2} dx$ put $\frac{x}{2} = t$

$$I = \int_0^{\frac{\pi}{2}} \sin^8 t 2dt \; dx = 2dt, x = 0, \pi \Rightarrow t = 0, \frac{\pi}{2}$$
$$= 2\left[\frac{7 \times 5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} \times \frac{\pi}{2}\right] = \frac{35\pi}{128}.$$

7. $\int_0^{25} [\sqrt{x}]dx = \int_0^1 [\sqrt{x}]dx + \int_1^4 [\sqrt{x}]dx + \int_4^9 [\sqrt{x}]dx + \int_9^{16} [\sqrt{x}]dx + \int_{16}^{25} [\sqrt{x}]dx$

$= \int_0^1 0 \cdot dx + \int_1^4 1dx + \int_4^9 2dx + \int_9^{16} 3dx + \int_{16}^{25} 4dx$

$= 0 + (4-1) + 2(9-4) + 3(16-9) + 4(25-16) = 3 + 10 + 21 + 36 = 70.$

8. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \cdot \cos^6 x dx = 2\int_0^{\frac{\pi}{2}} \sin^4 x \cdot \cos^6 x \cdot dx = 2\left[\frac{3 \times 1 \times 5 \times 3 \times 1}{10 \times 8 \times 6 \times 4 \times 2} \times \frac{\pi}{2}\right] = \frac{3\pi}{256}.$

9. $Lt_{x \to 0} \frac{\int \cos^3 t \cdot dt}{x} = Lt_{x \to 0} \frac{\cos^3 x}{1} = 1.$

10. $\int_0^2 x^2 [x]dx = \int_0^1 0 + \int_1^2 x^2 dx = 0 + \left(\frac{x^3}{3}\right)_1^2 = \frac{7}{3}.$

11. $Lt_{n \to \infty} \left[\frac{1}{3n+1} + \frac{1}{3n+2} + \cdots \ldots \ldots + \frac{1}{3n+n}\right] \Rightarrow Lt_{n \to \infty} \sum_{r=1}^{n} \frac{1}{3n+r} \Rightarrow Lt_{n \to \infty} \frac{1}{n} \cdot \sum_{r=1}^{n} \frac{1}{3+\left(\frac{r}{n}\right)}$

$= \int_0^1 \frac{1}{3+x} dx = [\log(3+x)]_0^1 = \log 4 - \log 3 = \log\left(\frac{4}{3}\right).$

12. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta$ then $I_n + I_{n-2} = \frac{1}{n-1}$ subn $= 5 \Rightarrow I_5 + I_3 = \frac{1}{4} \Rightarrow \frac{1}{I_5 + I_3} = 4.$

13. $\int_0^{\pi} x \sin^3 x dx = \int_0^{\pi} (\pi - x) \sin^3 (\pi - x) dx = \int_0^{\pi} (\pi - x) \sin^3 x dx$

$I = \pi \int_0^{\pi} \sin^3 x dx - \int_0^{\pi} x \sin^3 x dx \Rightarrow 2I = \pi \int_0^{\pi} \sin^3 x dx \Rightarrow 2I = 2\pi \int_0^{\frac{\pi}{2}} \sin^3 x dx$

$I = \pi \int_0^{\frac{\pi}{2}} \frac{3\sin x - \sin 3x}{4} \cdot dx = \frac{\pi}{4}\left[-3\cos x + \frac{\cos 3x}{3}\right]_0^{\frac{\pi}{2}} = \frac{3\pi}{4} - \frac{\pi}{12} = \frac{2\pi}{3}.$

14. $Lt_{n \to \infty} \left[\frac{1+4+9+\cdots \ldots +n^2}{n^3}\right] \Rightarrow Lt_{n \to \infty} \sum_{r=1}^{n} \frac{r^2}{n^3} = Lt_{n \to \infty} \frac{1}{n} \cdot \sum_{r=1}^{n} \left(\frac{r}{n}\right)^2 = \int_0^1 x^2 dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$

15. $\underset{n \to \infty}{Lt} \sum_{r=1}^{n} \frac{1}{n} e^{\frac{r}{n}} = \int_0^1 e^x dx = e - 1.$

16. $I = \int_0^{\pi} [\cot x]dx = \int_0^{\pi} [\cot(\pi - x)]dx = \int_0^{\pi} [-\cot x]dx$

$2I = \int_0^{\pi} \{[\cot x] + [-\cot x]\}dx = \int_0^{\pi} (-1)dx \Rightarrow 2I = -\pi \Rightarrow I = \frac{-\pi}{2}.$

17. $\int_0^2[|x|+|x-1|]dx=\int_0^1[|x|+|x-1|]dx+\int_1^2[|x|+|x-1|]dx$

$=\int_0^1[x-(x-1)]dx+\int_1^2[x+(x-1)]dx=\int_0^1 1dx+\int_1^2(2x-1)dx$

$=[x]_0^1+[x^2-x]_1^2=1+(4-2)-(1-1)=3.$

18. $\underset{n\to\infty}{lt}\left[\frac{1^3}{n^4+1^4}+\frac{2^3}{n^4+2^4}+\cdots\ldots+\frac{1}{2n}\right]=\underset{n\to\infty}{lt}\sum_{r=1}^n\frac{r^3}{n^4+r^4}=\frac{1}{n}\cdot\sum_{r=1}^n\frac{\left(\frac{r}{n}\right)^3}{1+\left(\frac{r}{n}\right)^4}$

$=\int_0^1\frac{x^3}{1+x^4}dx=\frac{1}{4}\int_0^1\frac{4x^3}{1+x^4}dx=\frac{1}{4}[\log(1+x^4)]_0^1=\frac{1}{4}\log 2$

19. $\mathrm{lt}_{n\to\infty}\frac{\sqrt{1}+\sqrt{2}+\cdots\ldots+\sqrt{n}}{n\sqrt{n}}=\mathrm{lt}_{n\to\infty}\sum_{r=1}^n\frac{\sqrt{r}}{n\sqrt{n}}=\mathrm{lt}_{n\to\infty}\frac{1}{n}\cdot\sqrt{\frac{r}{n}}=\int_0^1\sqrt{x}\cdot dx=\left(\frac{2}{3}x^{3/2}\right)_0^1=\frac{2}{3}.$

20. $\sin^{-1}\left(\frac{x}{1+x^2}\right)$ is an odd function $\forall x\in(-1,1)\ \therefore\int_{-1}^1\sin^{-1}\left(\frac{x}{1+x^2}\right)dx=0.$

21. Let $y=\mathrm{lt}_{n\to\infty}\left[\left(1+\frac{1^2}{n^2}\right)\cdot\left(1+\frac{2^2}{n^2}\right)\ldots\ldots\ldots\left(1+\frac{n^2}{n^2}\right)\right]^{1/n}$

$\log y=\mathrm{lt}_{n\to\infty}\frac{1}{n}\left[\log\left(1+\frac{1^2}{n^2}\right)+\log\left(1+\frac{2^2}{n^2}\right)+\cdots..+\log\left(1+\frac{n^2}{n^2}\right)\right]=\underset{n\to\infty}{\frac{1}{n}}\sum_{r=1}^n\log\left(1+\frac{r^2}{n^2}\right)$

$=\int_0^1\log(1+x^2)dx=[x\cdot\log(1+x^2)]_0^1-\int_0^1\frac{2x^2}{1+x^2}=\log 2-2\int_0^1\left(1-\frac{1}{1+x^2}\right)dx$

$=\log 2-2[x-\mathrm{Tan}^{-1}x]_0^1=\log 2-2\left(1-\frac{\pi}{4}\right)=\log 2-2+\frac{\pi}{2}=\log 2+\log e^{\left(\frac{\pi}{2}-2\right)}$

$\log y=\log\left(2.e^{\left(\frac{\pi}{2}-2\right)}\right)\Rightarrow y=2\,e^{\left(\frac{\pi-4}{2}\right)}.$

22. $\Rightarrow\int_1^4|x-3|dx=\int_1^3|x-3|dx+\int_3^4|x-3|dx=\int_1^3(3-x)dx+\int_3^4(x-3)dx=\left(3x-\frac{x^2}{2}\right)_1^3+\left(\frac{x^2}{2}-3x\right)_3^4=2+\frac{1}{2}=\frac{5}{2}.$

23. Statement-I: $\int_\pi^{10\pi}|\sin x|dx=\int_0^{9\pi}|\sin x|dx=9\int_0^\pi|\sin x|dx\Rightarrow 9(2)=18.$

Statement-II: $\int_0^x x^2[x]dx=\int_0^1 0dx+\int_1^2 x^2=0+\left(\frac{x^3}{3}\right)_1^2=\frac{7}{3}.$

24. $\frac{7\times5\times3}{8\times6\times4\times2}\cdot\frac{\pi}{2}=\frac{35\pi}{256}.$ 25. $\frac{7\times5\times3}{8\times6\times4\times2}\cdot\frac{\pi}{2}=\frac{35\pi}{256}.$

25. $\frac{4\times2}{5\times3\times1}=\frac{8}{15}.$

(a) $I=4\int_0^{\frac{\pi}{2}}\sin^4 x\cos^6 xdx=4\frac{3\times1\times5\times3\times1}{10\times8\times6\times4\times2}\times\frac{\pi}{2}=\frac{3\pi}{128}.$

(b) $I=2\int_0^{\frac{\pi}{2}}\sin^4 x\cdot\cos^6 x\cdot dx=2\times\frac{3\pi}{512}=\frac{3\pi}{256}.$

(c) $=\int_0^{\frac{\pi}{6}}\cos^4 3\theta\sin^2 6\theta d\theta(put3\theta=x)=\frac{4}{3}\times\frac{1}{8}\times\frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}=\frac{5\pi}{192}.$

(d) $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^3 x\cos^2 xdx+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin^2\cdot\cos^3 xdx=2\int_0^{\frac{\pi}{2}}\sin^2 x\cdot\cos^3 xdx=2\times\frac{1\times2}{5\times3\times1}=\frac{4}{15}.$

28. $I = \int_0^\infty \frac{x^2+1-x^2}{1+x^4}dx = \int_0^\infty \frac{x^2}{1+x^4}dx + \int_0^\infty \frac{1-x^2}{1+x^4} = I_1 + I_2$

$I_2 = \int_0^\infty \frac{\frac{1}{x^2}-1}{x^2+\frac{1}{x^2}}dx$ put $x + \frac{1}{x} = y \Rightarrow I_2 = \int_\infty^\infty \frac{-1}{y^2-2}dy = 0 \therefore I = \int_0^\infty \frac{x^2}{1+x^4}dx.$

$$2I = \int_0^\infty \frac{1+x^2}{1+x^4}dx,\ 2I = \int_0^\infty \frac{\frac{1}{x^2}+1}{\frac{1}{x^2}+x^2}dx$$

Put $x - \frac{1}{x} = y, 2I = \int_{-\infty}^\infty \frac{1}{y^2+2}dy, = \left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{y}{\sqrt{2}}\right]_{-\infty}^\infty = \frac{\pi}{\sqrt{2}} \therefore I = \frac{\pi}{2\sqrt{2}}.$

29. (a) $\Rightarrow \int_0^\pi [\cos x]dx = \int_0^{\frac{\pi}{2}} 0dx + \int_{\frac{\pi}{2}}^\pi [\cos x]dx = 0 + \int_{\frac{\pi}{2}}^\pi [-1]dx \Rightarrow (-x)_{\frac{\pi}{2}}^\pi = -\pi + \frac{\pi}{2} = -\frac{\pi}{2}.$

(b) $\int_{-1}^1 (x - [2x])dx = -\int_{-1}^1 [2x]dx = -\left\{\int_{-1}^{-\frac{1}{2}} -2dx + \int_{-\frac{1}{2}}^0 -1dx + \int_0^{\frac{1}{2}} 0dx + \int_{-\frac{1}{2}}^1 1dx\right\}$

$= -\left\{-2\left(-\frac{1}{2}+1\right) - 1\left(\frac{1}{2}\right) + \left(1-\frac{1}{2}\right)\right\} = 1.$

c) $\int_0^3 (x - [x])dx = \int_0^1 (x - [x])dx + \int_1^2 (x - [x])dx + \int_1^3 (x - [x])dx$

$= \int_0^1 xdx + \int_1^2 (x-1)dx + \int_1^3 (x-2)dx$

$= \left(\frac{x^2}{2}\right)_0^1 + \left(\frac{x^2}{2} - x\right)_1^2 + \left(\frac{x^2}{2} - 2x\right)_2^3 = \frac{1}{2} + \left(0 + \frac{1}{2}\right) + \left(\frac{-3}{2} + 2\right) = \frac{3}{2}.$

(d) $\int_0^{\frac{\pi}{2}} \sqrt{1-\cos 2x}dx = \int_0^{\frac{\pi}{2}} \sqrt{2}\sin x dx = \sqrt{2}[-\cos x]_0^{\frac{\pi}{2}} = \sqrt{2}[0+1] = \sqrt{2}.$

30. $s_n = \sum_{k=0}^n \frac{n}{n^2+kn+k^2}$

$$s_n = \sum_{k=0}^n \frac{1}{n}\left(\frac{n}{1+\frac{k}{n}+\frac{k^2}{n^2}}\right) < \text{lt}_{n\to\infty}\frac{1}{n}\left(\frac{1}{1+\left(\frac{k}{n}\right)+\left(\frac{k}{n}\right)^2}\right) = \int_0^1 \frac{1}{1+x+x^2}dx$$

$$= \left[\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2}{\sqrt{3}}\left(x+\frac{1}{2}\right)\right)\right]_0^1$$

$= \frac{2}{\sqrt{3}}\left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}} \Rightarrow$ i.e., $s_n < \frac{\pi}{3\sqrt{3}}$ and $T_n > \frac{\pi}{3\sqrt{3}}.$

31. $\int_{-2}^2 |1-x^2|dx = 2\int_0^2 |1-x^2|dx = 2\int_0^1 |1-x^2|dx + 2\int_0^2 |1-x^2|dx = 2\left[x - \frac{x^3}{3}\right]_0^1 + 2\left[\frac{x^3}{3} - x\right]_1^2$

$= 2\left[1 - \frac{1}{3}\right] + 2\left[\frac{8}{3} - 2 - \frac{1}{3} + 1\right] = \frac{4}{3} + \frac{8}{3} = 4.$

32. (a) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx = \left(\frac{b-a}{2}\right) = \frac{\frac{\pi}{3}-\frac{\pi}{6}}{2} = \frac{\pi}{12}.$

(b) $I = \int_{\frac{\pi}{4}}^{3\frac{\pi}{4}} \frac{1}{1+\cos x}dx - - - - - - - - (1)\ I = \int_{\frac{\pi}{4}}^{3\frac{\pi}{4}} \frac{1}{1-\cos x}dx -$

$= \int_{\frac{\pi}{4}}^{3\frac{\pi}{4}} 2\text{cosec}\, e^2 x dx \Rightarrow I = (-\cot x)_{\frac{\pi}{4}}^{\frac{3\pi}{4}} = 2.$

(c) $I = \int_0^{\pi} [\cot x]dx = \int_0^{\pi} [\cot(\pi - x)]dx = \int_0^{\pi} [-\cot x]dx$

$\therefore 2I = \int_0^{\pi} [\cot x]dx + \int_0^{\pi} [-\cot x]dx \begin{bmatrix} \because [x] + [-x] = -1, \text{if } x \notin z \\ = 0, \text{if } x \in z \end{bmatrix} = \int_0^{\pi} (-1)dx = (-x)_0^{\pi} = -\pi \Rightarrow I = \frac{-\pi}{2}$.

d) $\int_{\frac{\pi}{2}}^{3\frac{\pi}{2}} [2\sin x]dx = \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} 1dx + \int_{\frac{5\pi}{6}}^{\pi} 0dx + \int_{\pi}^{\frac{7\pi}{6}} (-1)dx + \int_{\frac{7\pi}{6}}^{3\frac{\pi}{2}} (-2)dt = \frac{5\pi}{6} - \left(\frac{7\pi}{6} - \pi\right) - 2\left(\frac{3\pi}{2} - \frac{7\pi}{6}\right) = -\frac{\pi}{2}$.

# 6. Circles

## WORKSHEET – 1
## Level – I

1) A 2) C 3) D 4) A 5) B 6) A 7) C

8) B 9) D 10) D 11) A

1. Given eq.of circle is $x^2 + y^2 - 8x + 10y + 9 = 0 \Rightarrow 2g = -8, \Rightarrow g = -4 \Rightarrow 2f = 10$
$\Rightarrow f = 5$

$$\therefore \text{ centre } = (-g, -f) = (4, -5)$$

2. Given eq.of the circle $x^2 + y^2 + 6x + 8y - 39 = 0 \Rightarrow 2\,g = 6 \Rightarrow g = 3, 2f = 8 \Rightarrow f = 4,$
$c = -39 \Rightarrow \therefore$ radius $= r = \sqrt{g^2 + f^2 - c} = \sqrt{3^2 + 4^2 - (-39)} = \sqrt{64} = 8$ units

3. Given that centre $= (h, k) = (-1, -8)$; radius $= r = 5$ cm
$\therefore$ Required equation of circle $\Rightarrow (x - h)^2 + (y - k)^2 = r^2$

$(x + 1)^2 + (y + 8)^2 = 5^2 \Rightarrow x^2 + y^2 + 2x + 16y + 40 = 0$

4. Given the end points of the diameter of the circle are
$(x_1, y_1) = (1,2)$ and $(x_2, y_2) = (7,8)$

$\Rightarrow$ req.eq.of circle $\Rightarrow (x - x_1)(x - x_2) + (y - y_2)(y - y_2) = 0$

$\Rightarrow (x - 1)(x - 7) + (y - 2)(y - 8) = 0 \Rightarrow x^2 - 8x + 7 + y^2 - 10y + 16 = 0$

$\Rightarrow x^2 + y^2 - 8x - 10y + 23 = 0$

5. Intercept made by the circle on x axis $= \overline{AB} = 2\sqrt{g^2 - c}$
$\Rightarrow 2g = 10 \Rightarrow g = 5, 2f = 15, c = 16$
$\therefore \overline{AB} = 2\sqrt{g^2 - c} = 2 \times \sqrt{5^2 - 16} = 2 \times 3 = 6$

6. We know that intercept made by the circle on y-axis $\overline{AB} = 2\sqrt{f^2 - c}$
given: $x^2 + y^2 + 10x + 8y - 33 = 0 \Rightarrow 2g = 10, 2f = 8, c = -33 \ \therefore \overline{AB} = 2\sqrt{4^2 - (-33)}$

$= 2\sqrt{16 + 33} = 2 \times 7 = 14.$

7. Given the centre of circle $= (h, k) = (0,9) = A$ passing through orgin $= O = (0,0)$
$\therefore r = OA = \sqrt{(0 - 0) + (0 - 9)}^2 = 9$

8. Concept

9. given $x^2 + y^2 + 4x - 10y - 7 = 0$ and $P(4, -3)$

Centre $= (-g, -f) = (-2,5)$ and ; Radius $= \sqrt{g^2 + f^2 - c} = 6$

$CP = \sqrt{(4 + 2)^2 + (-3 - 5)^2} = 10$; Minimum distance $= |CP - r| = |10 - 6| = 4$

Maximum distance $= |CP + r| = |10 + 6| = 16$

Sum of Minimum and Maximum distance = 4 + 16 = 20.

10. Given $P(-3,2)$ and $x^2 + y^2 - 2x + 2y + 1 = 0$

$\Rightarrow$ Centre = $(1,-1)$ and Radius = $\sqrt{g^2 + f^2 - c} = 1 \Rightarrow CP = \sqrt{(1+3)^2 + (-1-2)^2} = 5$

Longest distance = $|CP + r| = |5 + 1| = 6$

11. Concept

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) A | 2) C | 3) D | 4) C | 5) A | 6) B | 7) B |
| 8) A | 9) C | 10) B | 11) C | 12) B | 13) B | 14) D |
| 15) A | 16) B | 17) B | 18) B | 19) A | 20) B | 21) C |
| 22) D | 23) C | 24) A,B,C,D | 25) A,B,C,D | 26) D | 27) A | 28) D |
| 29) a:r, b:s, c:p, d:q | | 30) 8 | 31) A,B,C,D | 32) A | 33) a:r, b:p, c:t, d:t | |
| 34) A | | | | | | |

1. Given equation of circle: $\Rightarrow 3x^2 + 3y^2 - 6x + 4y + k = 0 \Rightarrow x^2 + y^2 - 2x + \frac{4}{3}y + \frac{k}{3} = 0$

$\Rightarrow g = -1, f = \frac{2}{3}, c = \frac{k}{3}$ $\therefore$ given $r = \frac{5}{3} \Rightarrow \sqrt{g^2 + f^2 - c} = \frac{5}{3}$

$\Rightarrow \sqrt{(-1)^2 + \left(\frac{2}{3}\right)^2 - \frac{k}{3}} = \frac{5}{3} \Rightarrow k = -4$

2. Let the Equation of Circle be $x^2 + y^2 + 2gx + 2fy + c = 0$, then the Parametric Form is $x = -g + r\cos\theta, y = -f + r\sin\theta$.

Given $x^2 + y^2 + px + py = 0 \Rightarrow x^2 + 2x\left(\frac{p}{2}\right) + y^2 + 2y\left(\frac{p}{2}\right) = 0$

$\Rightarrow \left(x + \frac{p}{2}\right)^2 + \left(y + \frac{p}{2}\right)^2 = \frac{p^2}{2} = \left(\frac{p}{\sqrt{2}}\right)^2$

$\therefore$ The Parametric equations are $x = \frac{-p}{2} + \frac{p}{\sqrt{2}}\cos\theta,\ y = \frac{-p}{2} + \frac{p}{\sqrt{2}}\sin\theta$

3. We know that centroid divides median in 2:1 ratio also, we known that in equilateral triangle $0, C, S, I$ are coincide.

$\therefore$ Length of Altitude = Median = 9

$AD = 9 \therefore AG = \frac{2}{3} \times AD = \frac{2}{3} \times 9 = 6$ $\therefore$ Radius of the circle = $AG = 6$

$\Rightarrow r = 6$ and Centre = (0,0)

$\therefore$ Equationof circle is $x^2 + y^2 = 36$.

4. Given lines are $x - 2y + 3 = 0, 3x + ky + 7 = 0$

since the given lines are concyclic points $\Rightarrow a_1 a_2 = b_1 b_2$

$\Rightarrow 1 \times 3 = -2 \times k \ \Rightarrow\ k = \frac{-3}{2}$

5. Equation of the circle having r = 3 lies in $Q_4$ touching the lines x = 0, y = 0 is (h, k) = (3, −3)

⇒ Required equationof the circle ⇒ $(x-h)^2+(y-k)^2=r^2$

⇒ $(x-3)^2+(y+3)^2=3^2 \Rightarrow x^2+y^2-6x+6y+9=0$

6. Given equation of circle $x^2+y^2-14x-10y-151=0$

centre = (−g, −f) ⇒ g = −7, f = −5 ⇒ C = (7,5)

Radius = $\sqrt{49+25+151}=15$ point = (2, −7), C = (7,5) and

CP = $\sqrt{(2-7)^2+(-7-5)^2}=13$ ∴ The shortest distance

= $|CP-r|=|13-15|=2$ and the largest distance = $|CP+r|=|13+15|=28$

7. Given circle: $x^2+y^2-2x+8y-23=0$ point = (2,3)

we know that: value of $cp^2-r^2$ is power of a point w.r.t S = 0

⇒ $PA \times PB = S_{11} \Rightarrow cp^2-r^2=S_{11}$ ∴ Power of a point = $S_{11}$

⇒ $x_1^2+y_1^2-2x_1+8y-23=2^2+3^2-2(2)+8(3)-23=10.$

8. Equation of the circle concentric with the circle $x^2+y^2-6x-4y-12=0$ is $x^2+y^2-6x-4y+k=0$. It is passing through (−2,14)

∴ $(-2)^2+(14)^2-6(-2)-4(14)+k=0 \Rightarrow$ k = −156

∴ The required equation of circle is $x^2+y^2-6x-4y-156=0.$

9. Given the equation of circle $x^2+y^2=10$ ⇒ Centre = C = (0,0),

Radius = r = $\sqrt{10}$. Let d = distance from (0,0) to 3x − y + 4 = 0

= $\left|\frac{c}{\sqrt{1+m^2}}\right|$ (or) $\left|\frac{c}{\sqrt{a^2+b^2}}\right|=\frac{4}{\sqrt{10}} \Rightarrow$ r > d

⇒ The straight line intersects the circle at two distict points.

10. Equation of circle ⇒ $2x^2+2y^2-5x-4y-3=0 \Rightarrow S=x^2+y^2-\frac{5}{2}x-2y-\frac{3}{2}=0$
$S(4,2)=16+4-\frac{5}{2}\times 4-2\times 2-\frac{3}{2}=\frac{9}{2}>0 \Rightarrow$∴ (4,2) lies outside of the circle

11. Equation of circle

$x^2+y^2+2gx+2fy+c=0$, if it touches the $x$-axis $c=g^2$

(1,0) lies on of the circle ⇒ $1+2g+g^2=0$

$g=-1, c=1 \Rightarrow$ (2,3) lies on circle ⇒ $4+9+4g+6f+g^2=0 \Rightarrow f=\frac{-5}{3}$

Diameter = $2\sqrt{g^2+f^2-c}=\frac{10}{3}$

12. Point (1,0) lies on circle ⇒ $x^2+y^2+2x+4y-3=0$ $c(-1,-2)$

∴ Requred point $(2c-p)=(-3,-4)$

13. The circle passing through (2,0) and (0,4) having minimum radius is the, circle having (2,0) and (0,4) as ends of its diameters

∴ Equation of circle is

$(x-2)(x-0)+(y-0)(y-4)=0; x^2+y^2-2x-4y=0$

14. Required equation of the circle is $x^2+y^2-(-6)x-(4)y=0; x^2+y^2+6x-4y=0$

15. Let the equationof the circle be $x^2+y^2+2gx+2fy+c=0$

circle passes through $\left(m_i,\frac{1}{m_i}\right) \Rightarrow m_i^4+2gm_i^3+2fm_i{}^2+cm_i{}^2+1=0$ it's degree is 4

∴ Product of the roots $m_1m_2m_3m_4=\frac{1}{1}=1$

16. Given equation of circle: $x^2+y^2+10x-12y-56=0 \Rightarrow 2g=10 \Rightarrow g=5,\ c=-56$

$2f=-12 \Rightarrow$∴ Length of intercept made by x axis

$=2\sqrt{g^2-c}=2\sqrt{5^2+56}=2\sqrt{25+56}=18$

17. Point of intesection of lines $2x-3y=11$ and $3x-4y=18$, is the centre of the circle, $\pi r^2=154 \Rightarrow r=7$. Centre = (10,3)

Equation of circle $\Rightarrow x^2+y^2-20x-6y+60=0$

18. Given circle: $x^2+y^2-4x+6y-4=0, P=(1,1) \Rightarrow S_{11}=1+1-4+6-4=0$ ∴ P lies on the circle.

19. $s+\lambda L=0 \Rightarrow x^2+y^2+y-1+\lambda(x-y+1)=0$

centre $\left(\frac{-\lambda}{2},\frac{1-\lambda}{2}\right)$ lies on $x-y+1=0$

$\Rightarrow \lambda=\frac{3}{2}$, required eq.of circle is $2(x^2+y^2)+3x-y+1=0$

20. $x=\frac{2a(1-t^2)}{1+t^2}, y=\frac{4at}{1+t^2} \Rightarrow$ Put $t=\text{Tan}\frac{\theta}{2}$

$x=2a\text{Cos}\,\theta, y=2a\text{Sin}\,\theta \Rightarrow \frac{x}{2a}=\text{Cos}\,\theta, \frac{y}{2a}=\text{Sin}\,\theta$

$\frac{x^2}{4a^2}+\frac{y^2}{4a^2}=1 \Rightarrow x^2+y^2=4a^2 \Rightarrow r=2a$

21. Given equation of circle is $x^2+y^2+14x+6y-42=0$ ∴ $r=\sqrt{49+9+42}=10$.

Height of equilateral triangle which is inscribed in the circle $=\frac{3}{2}r=\frac{3}{2}\times 10=15$.

Side ' $a$ ' $=\frac{2h}{\sqrt{3}}=\frac{2\times 15}{\sqrt{3}}=\frac{30}{\sqrt{3}}$; Area of the triangle $=\frac{\sqrt{3}}{4}a^2=\frac{\sqrt{3}}{4}\times\frac{30}{\sqrt{3}}\times\frac{30}{\sqrt{3}}$

$=75\sqrt{3}$ sq. units.

22. $x^2+y^2-4x-4y-1=0 \Rightarrow$ centre $(2,2), r=\sqrt{4+4+1}=3$

∴ Number of points of the circle touches the rectangle = 3

23. Circle passes through the origin

(0,0) then c = 0

$2\sqrt{g^2-6}=4 \Rightarrow g=\pm 2$

$2\sqrt{f^2-c}=8 \Rightarrow f=\pm 4$

$\therefore$ The equation of required circle is $x^2+y^2\pm 4x\pm 8y=0$

24. Given $x\text{Cos}\,\alpha+y\text{Sin}\,\alpha=2 \Rightarrow \frac{x}{2\text{Sec}\,\alpha}+\frac{y}{2\text{Cosec}\,\alpha}=1$

$2\text{Sec}\,\alpha$ is + ve in $Q_4$ and $2\text{Cosec}\,\alpha$ is - ve in $Q_4$ value of g = $\left(\frac{g\text{Cos}\,\alpha+g\text{Sin}\,\alpha-2}{1}\right)$

$\therefore g=\frac{2}{\text{Cos}\,\alpha+\text{Sin}\,\alpha\pm 1}$ $\therefore$ value of $g=\left|\frac{g\text{Cos}\,\alpha-g\text{Sin}\,\alpha-2}{1}\right|$

$g=g(\text{Cos}\,\alpha-\text{Sin}\,\alpha)-2 \Rightarrow \therefore g=\dfrac{2}{\text{Cos}\,\alpha-\text{Sin}\,\alpha\pm 1}$

25. concept

26. Equation of circle is $x^2+y^2-a^2+\lambda(x\text{Cos}\,\alpha+y\text{Sin}\,\alpha-p)=0$

$\therefore$ C = $\left(\frac{-\lambda\text{Cos}\,\alpha}{2},\frac{-\lambda\text{Sin}\,\alpha}{2}\right) \therefore -\frac{\lambda}{2}\text{Cos}^2\,\alpha-\frac{\lambda}{2}\text{Sin}^2\,\alpha=p$

27. Equation of circle is $\Rightarrow x^2+y^2-a^2+\lambda(x\text{Cos}\,\alpha+y\text{Sin}\,\alpha-p)=0.$

28. C lies on $x\text{Cos}\,\alpha+y\text{Sin}\,\alpha=p$
$\Rightarrow -\frac{\lambda}{2}\text{Cos}^2\,\alpha-\frac{\lambda}{2}\text{Sin}^2\,\alpha=p \Rightarrow \lambda=-2\text{p}$

29. a) The point of intersection of diameters is considered to be the centre.

$\text{x}+\text{y}=6, \text{x}+2\text{y}=4 \Rightarrow \text{P}(2,6), \text{C}(8,-2), r=\sqrt{36+64}=10$

b) Let the equation of circle be $\Rightarrow x^2+y^2+2gx+2fy+c=0$

if circle passes through (0,0), (1,0) and (0,1) then $c=0, g=\frac{1}{2} f=\frac{1}{2}$

Centre $\left(-\frac{1}{2},-\frac{1}{2}\right)$, Radius $r=\frac{1}{\sqrt{2}}$ $\therefore k=0,\frac{5}{13}$

c) Equation of circle is $\Rightarrow x^2+y^2+2x-3=0$ here $\text{C}=(-1,0)$

ordinate of point = $(-\sqrt{3})$

d) Equation of circle is $\Rightarrow x^2+y^2+4x-6y+9=0$, (k, 3) lies on circle then

$k^2+9+4k-18+9=0 \Rightarrow k^2+4k=0 \Rightarrow \text{k}=-4.$

30. Equation of circle $\Rightarrow x^2+y^2-6x-4y-12=0$ $\therefore$ C(3,2) and P(−2,14) $r=\sqrt{9+4+12}=5, CP=\sqrt{25+144}=13$ shortest distance $|CP-r|=|13-5|=8.$

31. Let the equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

(2,1) passes through (1) $\Rightarrow 4\,g + 2f + c = -5$

(5,5) passes through (1) $\Rightarrow 10\,g + 10f + c = -50$

$(-6,7)$ passes through (1) $\Rightarrow -12\,g + 14f + c = -85$

$(2) - (1) \Rightarrow 6g + 8f = -45$

(5) $\Rightarrow$ (3) $-$ (2) $\Rightarrow -22g + 4f = -35$

Solving (5) and (6) we have $g = \frac{1}{2}$ and $f = -6$

Substituting 'g' and ' f ' values in (2) we have $c = 5 \therefore g + f + c = \frac{1}{2} - 6 + 5 = -\frac{1}{2}$

32. Statement -1: Given circle is

$x^2 + y^2 - 4x - 2y - 20 = 0 \Rightarrow 2\,g = -4 \Rightarrow g = -2, 2f = -2 \Rightarrow f = -1$

Centre $C = (2,1)$

$$r = \sqrt{(-2)^2 + (-1)^2 + 20} = 5$$

Given point $P = (10,7)$

$$CP = \sqrt{8^2 + 6^2} = 10$$

$\therefore$ Shortest distance $= |CP - r| = |10 - 5| = 5$

$\therefore$ Largest distance $= |cr + r| = |10 + 5| = 15$

Statement -2 : True $\therefore$ both statements are true

33. a) Given $x^2 + y^2 - 12x + 14y + 11 = 0 \ \therefore 2\sqrt{g^2 - c} = 2\sqrt{36 - 11} = 10$

b) Given circle:

$$ax^2 + ay^2 + bx + dy + 2 = 0 \; x^2 + y^2 + \frac{b}{a}x + \frac{d}{a}y + \frac{2}{a} = 0$$

$2g = \frac{b}{a} \Rightarrow g = \frac{b}{2a}$ and $2f = \frac{d}{a} \Rightarrow f = \frac{d}{2a}$

$\therefore$ Centre $= (-g, -f) = \left(\frac{-b}{2a}, \frac{-d}{2a}\right) = (1,2) \Rightarrow 2a + b = 0, 4a + d = 0$

$$\therefore \frac{2b + 3d}{a} = \frac{2(-2a) + 3(-4a)}{a} = \frac{-4a - 12a}{a} = \frac{-16a}{a} = -16$$

c) Power of (2,3) w.r.t $\Rightarrow x^2 + y^2 - 8x + 2y - k = 0$ is 5

$S_{11} = x_1^2 + y_1^2 - 8x_1 + 2y_1 - k = 5 \Rightarrow 4 + 9 - 16 + 6 - k = 5 \Rightarrow 3 - k = 5 \Rightarrow k = 2$

d) If $(2,3), (0,2), (4,5)$ and $(0, t)$ are concyclic then $t =$

Equation of the circle with AB on diameter is $x^2 + y^2 - 2x - 5y + 6 = 0$

Equation to $AB \Rightarrow x - 2y + 4 = 0$

Equation of circle passing through A,B is

$x^2 + y^2 - 2x - 5y + 6 + \lambda(x - 2y + 4) = 0$

This passes throguh

$c(4,5) \Rightarrow 16 + 25 - 8 - 25 + 6 + \lambda(4 - 10 + 4) = 0 \Rightarrow 14 - 2\lambda = 0 \Rightarrow \lambda = 7$

Equation of the circle passing through A,B,C $x^2 + y^2 + 5x - 19y + 34 = 0$

now, $D(0, t)$ lies on this circle $\Rightarrow t^2 - 19t + 34 = 0 \Rightarrow t = 2$ or $\frac{17}{2}$.

34. Circle touches x axis $\Rightarrow g^2 = c$; let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$; Circle touches x axis at a $(\pm 3,0)$

$$\Rightarrow 9 \pm 3(2g) + c = 0 \Rightarrow 9 \pm 3(2g) + g^2 = 0 \Rightarrow g^2 \pm 6g + 9 = 0$$

$$\Rightarrow (g \pm 3)^2 = 0 \Rightarrow g \mp 3 \Rightarrow c = 9$$

Intercept on y axis as $2\sqrt{7}$ distance

$$\Rightarrow 2\sqrt{7} = 2\sqrt{f^2 - c} \Rightarrow f^2 - c = 7 \Rightarrow f^2 - 9 = 7 \Rightarrow f^2 = 16 \Rightarrow f = \pm 4$$

$\therefore$ The required equation of circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \Rightarrow x^2 + y^2 \mp 6x \pm 8y + 9 = 0$$

$\Rightarrow x^2 + y^2 - 6x + 8y + 9 = 0$ and $x^2 + y^2 + 6x - 8y + 9 = 0$.

# 7. Parabola and Ellipse

## WORKSHEET – 1

## Level – I

| | | | | | |
|---|---|---|---|---|---|
| 1 ) C | 2) B | 3) A | 4) B | 5) C | 6) A |
| 7) A | 8) A | 9) B | 10) C | | |

1. Definition

2. $h^2 = ab$

3. $y^2 = 18x \Rightarrow y^2 = 4ax \Rightarrow 4ax = 18x \Rightarrow 4a = 18 \Rightarrow a = \frac{18}{4} = \frac{9}{2}$

   $\therefore$ Focus S = (a, 0) = $\left(\frac{9}{2}, 0\right)$

4. A Parabola has no centre Length of Latus Rectum is 4a

5. Given equation of Parabola

   $x^2 + 8x + 12y + 4 = 0 \Rightarrow x^2 + 8x + 16 + 12y + 4 = 16 \Rightarrow (x+4)^2 = -12y + 16 - 4$

   $\Rightarrow (x+4)^2 = -12y + 12 \Rightarrow (x+4)^2 = -12(y-1)$

   Comparing with the Standard Form $(x-h)^2 = 4a(y-k)$, we have

   Vertex = (h, k) = $(-4,1)$.

7. Eccentricity of the Parabola is always equal to ' 1 ' therefore value of $b$ is Real Number

8. $\frac{SP}{PM} = e = 1 \Rightarrow SP = PM \Rightarrow SP^2 = PM^2 \Rightarrow (x-1)^2 + y^2 = \frac{(x+2y-1)^2}{x^2+y^2}$

   $\Rightarrow 5(x^2 - 2x + 1 + y^2) = (x + 2y - 1)^2 \Rightarrow 5x^2 - 10x + 5 + 5y^2$
   $= x^2 + 4y^2 + 1 + 4xy - 4y - 2x$

   $\Rightarrow 4x^2 - 4xy + y^2 - 8x + 4y + 4 = 0$

9. Vertex(h, k) = $(-2,1)$

   Latus Rectum = $4a = 6$, Axis of Parabola is parallel to y-axis

   $\Rightarrow (x-h)^2 = 4a(y-k)$

   $\Rightarrow (x-(-2))^2 = 6(y-1) \Rightarrow (x+2)^2 = 6(y-1)$.

10. The Parabola $y = ax^2 + bx + c$ is symmetric about $x = \frac{-b}{2a}$

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) A | 3) B | 4) A | 5) A | 6) C | 7) C |
| 8) B | 9) A | 10) C | 11) C | 12) D | 13) D | 14) C |
| 15) A | 16) A | 17) B | 18) B | 19) A | 20) A | 21) B |

22) C 23) A,C,D 24) A,B,C,D 25) D 26) A 27) A

28) a-q,b-s,c-p,d-r 29) 8 30)A 31) B 32) C

33) B 34) a-s,b-p,c-q,d-r 35) C, D

1. Given the Equation of Parabola is $y^2 = 32x$.

   Comparing with Standard Form we have $4a = 32 \Rightarrow a = 8$

   If $S = y^2 - 32x$, then $S_{11} = y_1^2 - 32x_1$.

   Substituting $(8,-4)$ in $S_{11}$, we have $S_{11} = (-4)^2 - 32(8) = 16 - 256 = -240$

   $\therefore S_{11} < 0 \Rightarrow (8,-4)$ lies inside the Parabola.

2. Given $L = (-3,1), L^1 = (1,1)$

   Length of Latus Rectum

   $$LL^1 = 4a \Rightarrow \sqrt{(-3-1)^2 + (1-1)^2} = 4a \Rightarrow 4 = 4a \Rightarrow a = 1$$

   Focus (S) is the midpoint of Latus Rectum $S = \left(\frac{-3+1}{2}, \frac{1+1}{2}\right) = (-1,1) = (h, k+a)$

   $$\Rightarrow h = -1, k + a = 1 \Rightarrow k = 0$$

   Parabola is vertical, since Latus Rectum lies on $y = 1$ $\therefore$ Vertex$(h,k) = (-1,0)$

   $\therefore$ Equation of Parabola is of the form $(x-h)^2 = 4a(y-k)$

   $\Rightarrow (x-(-1))^2 = 4 \times 1(y-0) \Rightarrow (x+1)^2 = 4y$.

3. Given Focus $= (3,2)$ and Vertex $= (-1,2)$

   Here the y-coordinates are same $\Rightarrow y = 2$

   and the axis of Parabola is parallel to x-axis.

   $\therefore$ Equation of Parabola is of the form $(y-k)^2 = 4a(x-h)$

   $\therefore$ Focus $= (h+a, k) = (3,2)$ and Vertex $= (h,k) = (-1,2)$

   On comparing we have $h = -1, k = 2, a = 4$

   $\therefore$ Equation of the Parabola is $(y-2)^2 = 4 \times 4(x-(-1)) \Rightarrow (y-2)^2 = 16(x+1)$.

4. Given Focus $= S = (3,-4)$ and Directrix is $x + y + 7 = 0$. Let $P = (x,y)$

   We have $SP^2 = PM^2 \Rightarrow (x-3)^2 + (y+4)^2 = \frac{(x+y+7)^2}{1^2+1^2}$

   $$2[x^2 + y^2 - 6x + 8y + 25] = x^2 + y^2 + 49 + 2xy + 14y + 14x$$
   $$\Rightarrow x^2 + y^2 - 2xy - 26x + 2y + 1 = 0$$

5. We know that, Length of the Chord intercepted by the Parabola $y^2 = 4ax$ on the line $y = mx + c$ is $\frac{4}{m^2} \times \sqrt{a(a-mc)(1+m^2)}$

   given line is $y = 4x + 2 \Rightarrow m = 4, c = 2$

also $y^2 = 68x \Rightarrow 4a = 68 \Rightarrow a = 17$
$\therefore$ Length of the chord

$$= \frac{4}{m^2} \times \sqrt{a(a - mc)(1 + m^2)} = \frac{4}{4^2} \times \sqrt{17(17 - 4 \times 2)(1 + 4^2)} = \frac{1}{4} \times 17 \times 3 = \frac{51}{4}$$

6. Given the Equation of Parabola is $y^2 = 8x$.

Comparing with Standard Form we have $4a = 8 \Rightarrow a = 2$.

$\therefore$ Focus $= S = (a, 0) = (2,0)$.

$\therefore$ Substituting $(2, y)$ in $y^2 = 8x \Rightarrow y = \pm 4$.

$\therefore$ The required co-ordinates of the points on $y^2 = 8x$ are $(2,4)$ and $(2, -4)$.

7. If ' $\theta$ ' is the angle subtended by the Double Ordinate at Vertex of $y^2 = 4ax$ then the Length of the Double Ordinate $= 8a\text{Cot}\frac{\theta}{2}$.

We have Length of Double Ordinate $8a = 8a\text{Cot}\frac{\theta}{2} \Rightarrow \text{Cot}\frac{\theta}{2} = 1 \Rightarrow \theta = 90°$.

8. Given $x = t^2 + 1, y = 2t + 1 \Rightarrow t = \frac{y-1}{2}$ then we have $x = \left(\frac{y-1}{2}\right)^2 + 1$

$\Rightarrow (y - 1)^2 = 4(x - 1)$.

Comparing with Standard Form $Y^2 = 4aX$, here $Y = y - 1$ and $X = x - 1$ and

$a = 1$.

Equation of Directrix is $X = -a \Rightarrow x - 1 = -1 \Rightarrow x = 0$

9. Given equation of Parabola is $y^2 - 2x - 6y - 5 = 0$

$\Rightarrow y^2 - 6y + 9 - 2x - 5 - 9 = 0 \Rightarrow (y - 3)^2 = 2x + 14 = 2(x - (-7))$

$\Rightarrow (y - 3)^2 = 4 \times \frac{1}{2}(x - (-7)) \Rightarrow (y - k)^2 = 49(x - h)$

$\therefore$ Equation of Directrix is $x = h - ae = -7 - \frac{1}{2} = \frac{-15}{2} \Rightarrow 2x + 15 = 0$.

10. Given Equation of Parabola $y^2 = 8x$, Comparing with Standard Form, $a = 2$ Given Line $y = x + 4$, Comparing with $y = mx + c$, we have $m = 1, c = 4 \Rightarrow a < mc$. $\therefore$ line neither touches nor intersects the Parabola.

11. Since the Semi-Latus Rectum of a Parabola is the Harmonic Mean between the segments of any focal chord of Parabola.

SP, 4, SQ are in H.P. $\Rightarrow 4 = \frac{2 \times SP \times SQ}{SP + SQ}$

(Since If a, b, c ....are in H.P., then

$b = \frac{2ac}{a+c} \Rightarrow 4 = \frac{2 \times 6 \times SQ}{6 + SQ} \Rightarrow SQ = 3$.

12. If $(x_1, y_1), (x_2, y_2)$ are the extremities of the Focal Chord of $y^2 = 4ax$ then $x_1 x_2 = a^2$ and $y_1 y_2 = -4a^2$ then $x_1 x_2 + y_1 y_2 = -4a^2 + a^2 = -3a^2$

13. Any line passing through the Focus other than the Axis always meets the P a rabola in two distinct points. Hence, $m \in R - \{0\}$

14. Given $(\alpha, 1)$ lies inside the curves $x^2 + y^2 - 4 = 0, y^2 = 4x$

$S_{11} < 0 \Rightarrow \alpha^2 + 1 - 4 < 0 \Rightarrow \alpha^2 < 3 \Rightarrow |\alpha| < \sqrt{3}$ and $1 - 4\alpha < 0 \Rightarrow \alpha > \frac{1}{4}$

15. $L_1 = \ell_1 x + m_1 y + n_1 = 0 \Rightarrow 2x + 3y + 12 = 0$

$L_2 = \ell_2 x + m_2 y + n_2 = 0 \Rightarrow x - y + 4k = 0$

given $y^2 = 8x \Rightarrow 4a = 8 \Rightarrow a = 2$.

If two lines are Conjugate w.r.t. $y^2 = 4ax$ then

$y = x^2 + 2px + 13 \Rightarrow y = (x + p)^2 + 13 - p^2 \Rightarrow (x + p)^2 = y - (13 - p^2)$

$(x - (-p))^2 = 4 \times \frac{1}{4}\left(y - (13 - p^2)\right) \Rightarrow 2(4k) + 1(12) = 2 \times 2 \times 3(-1) \Rightarrow 8k + 12 = -12$

$\Rightarrow 8k = -24 \Rightarrow k = -3$

16. Given that the Parabola $y = ax^2 + bx + c$ passes through $(0,3), (1,-4)$ and $(-1,4)$. Substituting these points in the equation, we have:

$c = 3, a + b + c = -4$ and $a - b + c = 4$.

Simplifying we have $a + b = -7$ and $a - b = 1$

Solving we have $a = -3$ and $b = -4$

Substituting $a, b, c$ values in $y = ax^2 + bx + c$ we have $y = -3x^2 - 4x + 3$.

17. The Vertex of the Parabola $\Rightarrow (y - k)^2 = 4a(x - h)$ or $(x - h)^2 = 4a(y - k)$ is $(h, k)$. The given Parabol is $y = x^2 + 2px + 13 \Rightarrow y = (x + p)^2 + 13 - p^2 \Rightarrow (x + p)^2 = y - (13 - p^2)$

$$(x - (-p))^2 = 4 \times \frac{1}{4}\left(y - (13 - p^2)\right) \therefore \text{ Vertex } = (-p, 13 - p^2)$$

Also given that the vertex is 4 units above the x-axis y-co ordinate of the vertex is 4 .

$13 - p^2 = 4, p^2 = 13 - 4, p^2 = 9, p = \pm 3$

18. The co-ordinates of the Focus and Vertex of the required Parabola are $S(a_1, 0)$ and $A(a, o)$. $\therefore$ the distance between the Vertex and Focus is $AS = (a_1 - a)$ and so the Length of the Latus Rectum $= 4(a_1 - a)$.
$\therefore$ the equation of the Parabola is $y^2 = 4(a_1 - a)(x - a)$.

19. Given $x = t^2 - t + 1$ and $y = t^2 + t + 1 \Rightarrow x + y = 2(t^2 + 1), y - x = 2t$

$\Rightarrow t = \frac{y-x}{2}$ Substituting ' t ' value $\Rightarrow \frac{x+y}{2} = \left(\frac{y-x}{2}\right)^2 + 1 \Rightarrow (y - x)^2 = 2(x + y) - 4$

$\Rightarrow (y - x)^2 = 2(x + y - 2)$

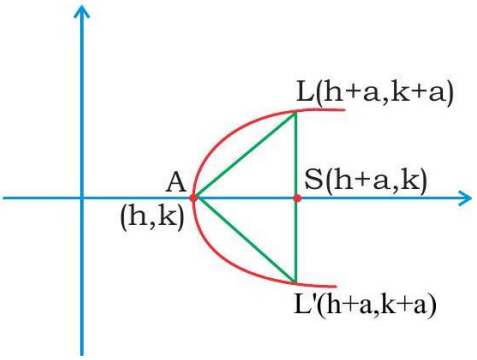
$\therefore$ Vertex will be the point where lines $y - x = 0$ and $x + y - 2 = 0$ meet $= (1,1)$.

20. Given Ends of Latus Rectum are

$L(h + a, k + 2a) = (1,3) \Rightarrow h + a = 1, k + 2a = 3$ and

$L^1(\mathrm{h}+\mathrm{a},\mathrm{k}-2\mathrm{a})=(1,-1)\Rightarrow \mathrm{h}+\mathrm{a}=1,\mathrm{k}-2\mathrm{a}=-1.$

Solving the equations we have $\mathrm{k}=1,\mathrm{a}=1,\ \mathrm{h}=0.$




$\Rightarrow$ Vertex $A=(\mathrm{h},\mathrm{k})=(0,1);\therefore \Delta ALL^1=\frac{1}{2}\begin{vmatrix}0 & 1 & 1 & 0\\ 1 & 3 & -1 & 1\end{vmatrix}=\frac{1}{2}|-1-1-3+1|=2$ sq. units.

21. Given Parabola $x^2=6y\Rightarrow$ Focus $\Rightarrow S=\left(0,\frac{3}{2}\right),L=\left(3,\frac{3}{2}\right)$ and $L^1=\left(-3,\frac{3}{2}\right).$

$\therefore\ LL^1=6$, and $LL^1$ is parallel to $x$-axis.

Equation to $\overrightarrow{\mathrm{OL}}$ is $\mathrm{y}=\frac{1}{2}\mathrm{x}$ and Equation to $\overrightarrow{\mathrm{OL^1}}$ is $\mathrm{y}=-\frac{1}{2}\mathrm{x}$

Combined equation of $\overrightarrow{\mathrm{OL}}$&$\overrightarrow{\mathrm{OL^1}}$ is $\left(\mathrm{y}-\frac{1}{2}\mathrm{x}\right)\left(\mathrm{y}+\frac{1}{2}\mathrm{x}\right)=0\Rightarrow \mathrm{x}^2-4\mathrm{y}^2=0.$

22. Let $\mathrm{A}=(4,-6)$ and $\mathrm{B}=(3,1)$

Let the Parabola $\mathrm{y}^2=4\mathrm{x}$ divide $\overline{\mathrm{AB}}$ in the ratio $\mathrm{K}:1$ at P

$\therefore P=\left(\frac{3K+4}{K+1},\frac{K-6}{K+6}\right)$ and $P$ is a point on $y^2=4x\Rightarrow\left(\frac{K-6}{K+1}\right)^2=4\left(\frac{3K+4}{K+1}\right)$

$\Rightarrow(\mathrm{K}-6)^2=4(3\ \mathrm{K}+4)(\mathrm{K}+1)\Rightarrow 11\ \mathrm{K}^2+40\ \mathrm{K}-20=0$

$$K=\frac{-40\pm\sqrt{1600-4(11)(20)}}{2\times 11}$$

$=\frac{-40\pm\sqrt{2480}}{22}=\frac{-20\pm2\sqrt{155}}{11}$ Ratio $=\mathrm{K}:1=-20\pm2\sqrt{155}:11.$

23. Given Parabola $x^2+4y-6x+k=0\Rightarrow(x^2-6x+9)-9+4y+k=0$

$$\Rightarrow(x-3)^2=9-k-4y=4\left(y-\frac{9+k}{4}\right)\Rightarrow(x-3)^2=4(-1)\left(y-\left(\frac{9-k}{4}\right)\right)$$

Comparing with Standard Form $(x-h)^2=4a(y-k)$ we have $\mathrm{a}=-1.$

$\therefore$ Vertex $\mathrm{A}=(\mathrm{h},\mathrm{k})=\left(3,\frac{9-k}{4}\right)$ $\therefore$ Focus $\mathrm{S}=(\mathrm{h},\mathrm{a}+\mathrm{k})$

$$=\left(3,-1+\frac{9-k}{4}\right)=\left(3,\frac{5-k}{4}\right)$$

Given Equation of Directrix is $\mathrm{y}-18=0,\Rightarrow \mathrm{y}=18.$

Also we have the Equation of Directrix is $\mathrm{y}=\mathrm{k}-\mathrm{a}\Rightarrow 18=\mathrm{k}-(-1)\Rightarrow \mathrm{k}=17\ \therefore$

Focus $= \left(3, \frac{5-k}{4}\right) = \left(3, \frac{5-17}{4}\right). = (3,-3)$ ∴ Vertex $- \left(3, \frac{9-k}{4}\right) = \left(3, \frac{9-17}{4}\right) = (3,-2).$

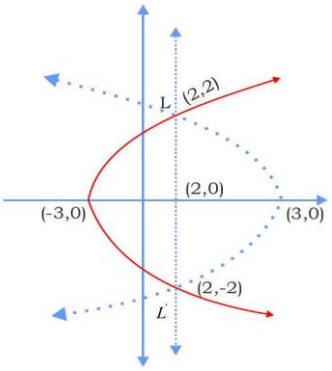
24. Required equation of Parabola $(y-k)^2 = \pm 4a(x-h)$

Verifying the options with the Vertices (h, k) = (1,0) and (3,0)

Considering Vertex A = (1,0) ⇒ h = 1, k = 0 then

Focus = S = (h + a, k) = (2,0) ⇒ h + a = 2 ⇒ a = 1,

then equation is $(y-0)^2 = 4(x-1) \Rightarrow y^2 = 4(x-1)$




Considering Vertex $A = (3,0) \Rightarrow h = 3, k = 0$ then

Focus = S = (h + a, k) = (2,0) ⇒ h + a = 2 ⇒ a = −1,

then equation is $(y-0)^2 = -4(x-3) \Rightarrow y^2 = -4(x-3)$.

25, 26, 27. We have $SP^2 = PM^2$

$$\Rightarrow (x+3)^2 + (y+4)^2 = \frac{(2x-3y-5)^2}{13}$$

∴ Focus S = (−3, −4) and Directrix is 2x − 3y − 5 = 0

The point of the intersection of the Axis and Directrix of the parabola is

$Z = \left(\frac{-41}{13}, \frac{-49}{13}\right)$ (Hint: $Z$ is the feet of the perpendicular from focus to directrix.)

28. Concept

29. Length of the Latus Rectum $= \left|\frac{\text{Coefficient of } y}{\text{Coefficient of } x^2}\right| = \left|\frac{-8}{1}\right| = 8.$

30. Given $(y-2)^2 = 6(x+3) \Rightarrow (\alpha, \beta) = (-3,2) \Rightarrow 4a = 6, a = \frac{3}{2}$ ∴ Focus $= \left(\frac{-3}{2}, 2\right)$

31,32,33. Given Vertex A = (3,2) and Focus S(2, −1)

A midpoint of SZ $\frac{h+2}{2} = 3, \frac{k-1}{2} = 2 \Rightarrow h = 4, k = 5 \Rightarrow Z = (4,5)$

Slope of SZ = 3, Equation of Directrix is $y - y_1 = m(x - x_1) \Rightarrow\Rightarrow$ x + 3y − 19 = 0.

It is in the form of $ax + by + c = 0$, Comparing we have $a = 1, b = 3,$

c = −19 then a + b + c = −15

Distance between Focus to Directrix = Perpendicular distance from Focus to the

Directrix = $\left|\frac{1(2)+3(-1)-19}{\sqrt{1^2+3^2}}\right| = 2\sqrt{10}$

34. a) Given $y^2 - x - 2y + 2 = 0 \Rightarrow y^2 - 2y + 1 - x + 1 = 0$

$\Rightarrow (y-1)^2 = x - 1 \Rightarrow (y-1)^2 = 4 \times \frac{1}{4}(x-1)$

On comparing with $(y-k)^2 = 4a(x-h)$ we have $\therefore$ Focus S = (h + a, k)

$= \left(1 + \frac{1}{4}, 1\right) = \left(\frac{5}{4}, 1\right)$

b) Given $y^2 - 8x - 4y - 4 = 0 \Rightarrow [y^2 - 2(2)y + 2^2] - 2^2 - 8x - 1 = 0 \Rightarrow (y-2)^2 = 8x + 4 + 1$

$\Rightarrow (y-2)^2 = 8x + 5 \Rightarrow (y-2)^2 = 8\left(x - \left(\frac{-5}{8}\right)\right)$.

Therefore the Focus S = (h + a, k) = (4,2).

c) Given $x^2 + 4x - 8y + 28 = 0 \Rightarrow (x+2)^2 = 8(y-3)$

$\therefore$ Focus S = (h, a + k) = $(-2, 2+3) = (-2,5)$

d) Given $x^2 - 2x - 8y - 23 = 0 \Rightarrow x^2 - 2x + 1 - 8y - 24 = 0 \Rightarrow (x-1)^2 = 8(y-(-3))$, a = 2

$\therefore$ Focus S = (h, a + k) = $(1, -1)$

35. Let the line x + y = 5 intersect the parabola $y = x^2 + x$ at A and B, solving them we have x = $1, -5$ $\therefore$ A(1,4), B(−5,10) and AB = $6\sqrt{2}$.

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## WORKSHEET – 2
## Level – I

| | | | | | |
|---|---|---|---|---|---|
| 1) C | 2) C | 3) D | 4) B | 5) C | 6) D |
| 7) B | 8) C | 9) A | 10) A | | |

1. Let $A(a,0), B(-a,o)$ and $P(x,y)$ substitute in $PA + PB = k$ and simplify $PA + PB = k$ represents an ellipse if $k > 2a, k > AB$ [$\therefore AB = 2a$]

2. Let the two given lines be taken as the co-ordinate axis.

Let $C(\alpha, \beta)$ be the centre of the ellipse in any position.

Here the position of the centre $C$ changes as the ellipse slides

Let ' $a$ ' and ' $b$ ' be the semi major and semi minor axis of the ellipse.

Eq. of the director circle of ellipse $(x-\alpha)^2 + (y-\beta)^2 = a^2 + b^2$

Since OX, OY are mutually perendicular and the equation (1) passes through (0,0) $\Rightarrow \alpha^2 + \beta^2 = a^2 + b^2$

Hence locus of $(\alpha, \beta)$ is $x^2 + y^2 = a^2 + b^2$

3. Eccentricity of rectangular hyperbola is $\sqrt{2}$

4. If $e_1, e_2$ are the eccentricities of the hyperbola and its conjugate then $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$

5. Centre of the ellipse is the point of the intersection of the lines $2x - 3y - 1 = 0$, $3x + 2y - 8 = 0$. Solving both the equations we have Centre = (2,1)

6. Here a < b, ∴ Foci $S, S^1 = \left(h, k \pm \sqrt{b^2 - a^2}\right) = (1{,}5 \pm \sqrt{9-5}) = (1,7), (1,3)$.

7. Standard form of $9x^2 + 25y^2 - 18x - 100y - 116 = 0$ can be

$$9(x^2 - 2x + 1) - 9 + 25(y^2 - 4y + 4) - 100 - 166 = 0 \Rightarrow \frac{9(x-1)^2}{225} + \frac{25(y-2)^2}{225} = \frac{225}{225}$$

$$\Rightarrow \frac{(x-1)^2}{25} + \frac{(y-2)^2}{9} = 1 \Rightarrow a^2 = 25 \Rightarrow a = 5, b^2 = 9 \Rightarrow b = 3 \Rightarrow a > b$$

∴ Length of Latus Rectum $= \frac{2b^2}{a} = \frac{2\times 9}{5} = \frac{18}{5}$.

8. Given $\frac{x^2}{36} + \frac{y^2}{9} = 1 \Rightarrow a^2 = 36 \Rightarrow a = 6$ and $b^2 = 9 \Rightarrow b = 3$

$$a > b \Rightarrow SP + S^1P = 2a = 2(6) = 12$$

9. Given equation is $9x^2 - 16y^2 - 36x + 96y - 252 = 0$

$9(x^2 - 4x + 4) - 36 - 16 \Rightarrow (y^2 + 6y + 9) + 144 - 252 = 0$

$9(x-2)^2 - 16(y+3)^2 = 252 + 36 - 144 = 288 - 144 = 144$

∴ Standard Equation is $\frac{(x-2)^2}{16} - \frac{(y+3)^2}{9} = 1$

∴ Vertices $(h \pm a, b) = (2 \pm 4{,}3) = (6,3)$ and $(-2,3)$.

10. Given equation is $\Rightarrow 9x^2 - 16y^2 - 18x - 32y - 151 = 0$

$9(x^2 - 2x + 1) - 9 - 16(y^2 + 2y + 1) + 16 - 151 = 0 \Rightarrow 9(x-1)^2 - 16(y+1)^2 = 151 + 9 - 16$

$= 160 - 16 = 144$

∴ Standard Equation is $\Rightarrow \frac{(x-1)^2}{16} - \frac{(y+1)^2}{9} = 1$

On comparing we have

$a^2 = 16 \Rightarrow a = 4 \Rightarrow$ Length of transverse axis $= 2a = 8$

## Level – II

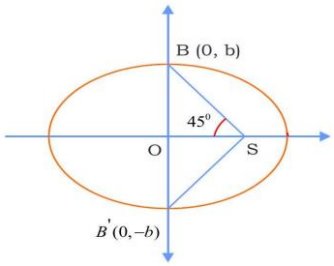
| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) A | 3) C | 4) A | 5) A | 6) C | 7) C |
| 8) B | 9) A | 10) C | 11) C | 12) D | 13) D | 14) C |
| 15) A | 16) A | 17) B | 18) B | 19) A | 20) A | 21) B |
| 22) C | 23) A,C,D | | | | | |

24) A,B,C,D 25) D 26) A 27) A 28) a-q,b-s,c-p,d-r

29) 8 30)A 31) B 32) C 33) B 34) a-s,b-p,c-q,d-r

1. $2a = 4(2b) \Rightarrow a = 4b$, Now $e^2 = \frac{a^2-b^2}{a^2} = \frac{15b^2}{16b^2} \therefore e = \frac{\sqrt{15}}{4}$

2. From $\triangle BOS$




$\text{Tan}\,45° = \frac{OB}{OS} = 1$; Here $OB = OS$

$$b = ae \Rightarrow b^2 = a^2e^2 \Rightarrow a^2(1-e^2) = a^2e^2 \Rightarrow 1 - e^2 = e^2 \Rightarrow 1 = 2e^2 \Rightarrow e = \frac{1}{\sqrt{2}}$$

3. Ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1 (a > b)$, On comparing with Standard form we have

$$a^2 = 9, b^2 = 4, e = \sqrt{\frac{a^2-b^2}{a^2}} = \sqrt{\frac{9-4}{9}} = \frac{\sqrt{5}}{3}$$

Ratio $= (1 + \text{e}):(1 - \text{e}) = 1 + \frac{\sqrt{5}}{3} : 1 - \frac{\sqrt{5}}{3} = (3 + \sqrt{5}):(3 - \sqrt{5})$

4. Given Ellipse is $\frac{x^2}{16} + \frac{y^2}{9} = 1$, On comparing $a^2 = 16 \Rightarrow a = 4$ and $b^2 = 9 \Rightarrow b = 3$

As a > b, we have $e = \sqrt{\frac{a^2-b^2}{a^2}} = \sqrt{\frac{7}{16}} = \sqrt{\frac{7}{4}}$

$\therefore$ the Focal distances $= a \pm ex_1 = 4 \pm \frac{\sqrt{7}}{4} \times 4 = 4 \pm \sqrt{7}$

5. Given $e = \frac{\sqrt{5}}{2}$, then

Length of latus rectum $= \frac{2b^2}{a} = \frac{2a^2(e^2-1)}{a} = 2a(e^2 - 1) = 2a\left(\frac{5}{4} - 1\right) = \frac{1}{4}(2a)$

6. Length of Latus rectum $= \frac{2b^2}{a} = \frac{22}{5} \Rightarrow \frac{2k}{5} = \frac{22}{5} \Rightarrow k = 11$

$$e = \sqrt{\frac{a^2+b^2}{b^2}} = \sqrt{\frac{11+25}{25}} = \frac{6}{5}$$

7. $e^2 = \frac{16+9}{16} = \frac{25}{16} \Rightarrow e = \frac{5}{4}$

8. Vertices are $A(c, c), A^1(-c, -c)$. $\therefore$ The equation of Transverse axis $\Rightarrow y = x$ Given Equation of Rectangular Hyperbola is $xy = 18 \Rightarrow xy = c^2$

$$\therefore c^2 = 18 \Rightarrow c = \sqrt{18} = 3\sqrt{2}$$

Length of the Transverse Axis

$= AA^1 = \sqrt{(2c)^2 + (-2c)^2} = \sqrt{8c^2} = c\sqrt{2} = 2(3\sqrt{2})\sqrt{2} = 12$

9. Given $9x^2 - 16y^2 - 72x - 32y - 16 = 0$, Converting into Standard Form

$9(x^2 - 8x + 16) - 144 - 16(y^2 + 2y - 1) + 14 - 16 = 0$

$\Rightarrow 9(x - y)^2 - 16(y + 1)^2 = 144 \Rightarrow \frac{(x-4)^2}{16} - \frac{(y+1)^2}{9} = 1$

Length of Latus Rectum $= \frac{2b^2}{9} = - = \frac{9}{2}$

10. Since a < b, Length of Major axis = 2 b = 6

11. Here $10 - a > 0 \Rightarrow a - 10 < 0 \Rightarrow a < 10$ and $4 - a > 0 \Rightarrow a - 4 < 0 \Rightarrow a < 4$

12. Given $3x^2 + 4y^2 + 12x - 8y - 32 = 0 \Rightarrow 3(x^2 + 4x + 4) - 12 + 4(y^2 - 2y + 1) - 4 - 32 = 0$

$3(x + 2)^2 + 4(y - 1)^2 - 12 + 4 + 32 = 48 \; \frac{(x+2)^2}{16} + \frac{(y-1)^2}{12} = 1$

Comparing with Standard Form $\frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$

we have $b^2 = a^2(1 - e^2) \Rightarrow 12 = 16(1 - e^2) \Rightarrow e^2 = 1 - \frac{12}{16} \Rightarrow e = \frac{1}{2}$

Equation of directrices is $x = \alpha \pm \frac{a}{e} \Rightarrow x = -2 \pm \frac{4}{\left(\frac{1}{2}\right)} = -2 \pm 8 \Rightarrow$ x = 6 and

x = −10

13. Given S = (0,0), $e = \sqrt{2}$ and Directrix is x + y + 1 = 0.

Let P(x, y) be any point on the Hyperbola

$\therefore$ SP = e(PM) $\Rightarrow SP^2 = e^2 \times PM^2 \Rightarrow (x - 0)^2 + (y - 0)^2 = 2\left|\frac{(x+y+1)}{\sqrt{2}}\right|^2$

$\Rightarrow x^2 + y^2 = x^2 + y^2 + 1 + 2xy + 2y + 2x \Rightarrow 2xy + 2x + 2y + 1 = 0$

14. Given $x^2 - 3y^2 - 4x - 6y - 11 = 0 \Rightarrow (x^2 - 4x + 4) - 3(y^2 + 2y + 1) = 12$

$\Rightarrow (x - 2)^2 - 3(y + 1)^2 = 12 \Rightarrow \frac{(x-2)^2}{12} - \frac{(y+1)^2}{4} = 1$

On comparing with Standard form we have

$a^2 = 12, b^2 = 4. \therefore$ We have eccentricity $e = \sqrt{\frac{b^2+a^2}{a^2}} = \frac{4}{\sqrt{3}}$

$\therefore$ Distance between the Foci $= 2ae = 2 \times \sqrt{12} \times \frac{2}{\sqrt{3}} = 8$

15. Given $\frac{2a}{2b} = \frac{2}{1} \Rightarrow \frac{a}{b} = 2 \Rightarrow a = 2b$

Area of rectangle $= (2a)(2b) = 4ab = 4(2b)b = 8b^2$

Verifying Options $8\left(\frac{4b^2+b^2}{5}\right) = 8\left(\frac{5b^2}{5}\right) = 8b^2$

16.. $Z = x + iy \Rightarrow Z^2 = (x+iy)^2 = x^2 - y^2 + 2xy \ \therefore \text{Im}(Z^2) = 32$

$\Rightarrow 2xy = 32 \Rightarrow xy = 16 \Rightarrow xy = c^2$

$\therefore$ It is a Rectangular Hyperbola and its ecentricity $= \sqrt{2}$.

17. We have $b^2 = a^2(e^2-1) \Rightarrow 4 = 9(e^2-1) \Rightarrow e^2 = \frac{13}{9} \Rightarrow e = \frac{\sqrt{13}}{3}$

18. We have $2ae = 16 \Rightarrow ae = 8$ given $e = \sqrt{2} \Rightarrow a = \frac{8}{\sqrt{2}} = 4\sqrt{2} \Rightarrow x^2 - y^2 = 32$

19. One extremity of the Latus rectum is

$$\text{L} = \left(ae, \frac{b^2}{a}\right) = (a\text{Cos}\,\theta, b\text{Sin}\,\theta) \Rightarrow \frac{b\text{Sin}\,\theta}{a\text{Cos}\,\theta} = \frac{b^2}{a(ae)}$$

$$\Rightarrow \text{Tan}\,\theta = \frac{b}{ae} \Rightarrow \theta = \text{Tan}^{-1}\left(\frac{b}{ae}\right)$$

20. We have $m_{AP} \times m_{AQ} = -1$ [$\because AP \perp AQ$]

$$\Rightarrow \frac{b\sin\alpha}{a\cos\alpha - a} \times \frac{b\sin\beta}{a\cos\beta - a} = -1$$

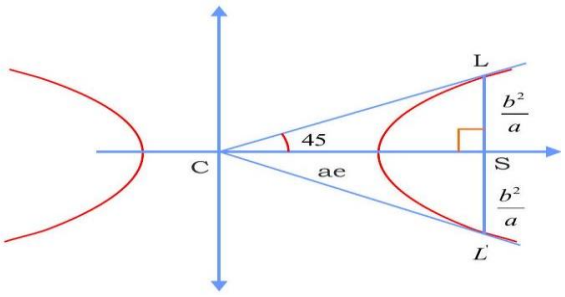
here $P(\alpha) = (a\cos\alpha, b\sin\alpha)$ and

$$Q(\beta) = (a\cos\beta, b\sin\beta) \Rightarrow \frac{b^2\sin\alpha\sin\beta}{a^2(\cos\alpha - 1)(\cos\beta - 1)} = -1$$

$$\Rightarrow \frac{b^2\left(2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\right)\left(2\sin\frac{\beta}{2}\cos\frac{\beta}{2}\right)}{a^2\left(1 - 2\sin^2\frac{\alpha}{2} - 1\right)\left(1 - 2\sin^2\frac{\beta}{2} - 1\right)} = -1 \Rightarrow \frac{4b^2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\sin\frac{\beta}{2}\cos\frac{\beta}{2}}{4a^2 \cdot \sin^2\frac{\alpha}{2} \cdot \sin^2\frac{\beta}{2}} = -1$$

$$\Rightarrow \frac{b^2\cos\alpha/2\cos\beta/2}{a^2\sin\alpha/2\sin\beta/2} = -1 \Rightarrow \tan\left(\frac{\alpha}{2}\right) \cdot \tan\left(\frac{\beta}{2}\right) = \frac{-b^2}{a^2}$$

21.




From $\Delta cs_2, \tan 45° = \frac{b^2/a}{ae}$

$$\Rightarrow \frac{b^2}{a} = ae \Rightarrow b^2 = a^2e \Rightarrow a^2(e^2-1) = a^2e[b^2 = a^2(e^2-1)] \Rightarrow e^2 - 1 = e \Rightarrow e^2 - e - 1 = 0$$

$$\therefore e = \frac{1 \pm \sqrt{1+4}}{2} \Rightarrow e = \frac{1 \pm \sqrt{5}}{2} \Rightarrow e = \frac{1+\sqrt{5}}{2}$$

22. Given $9(x^2+2x)-16(y^2-2y)=151 \Rightarrow 9(x+1)^2-16(y-1)^2=144$

$\Rightarrow \frac{(x+1)^2}{16}-\frac{(y-1)^2}{9}=1 \Rightarrow e=\frac{5}{4}$ $\therefore$ Foci $=(\alpha \pm ae, \beta)=(4,1),(-6,1)$.

23. Centre $A=(0,0)$ Point P on ellipse $P(\theta)=(a\cos\theta, b\sin\theta)$

Given $x^2+3y^2=6 \Rightarrow \frac{x^2}{6}+\frac{y^2}{2}=1 \Rightarrow a^2=6$ and $b^2=2$

Given AP $=2 \Rightarrow AP^2=4 \Rightarrow a^2\cos^2\theta+b^2\sin^2\theta=4 \Rightarrow 6\cos^2\theta+2\sin^2\theta=4$

$\Rightarrow 6\cos^2\theta+2(1-\cos^2\theta)-4=0 \Rightarrow 4\cos^2\theta-2=0$

$\Rightarrow \cos^2\theta=\frac{1}{2} \Rightarrow \cos\theta=\frac{1}{\sqrt{2}} \Rightarrow \cos\frac{\pi}{4}=\cos\frac{3\pi}{4}=\cos\frac{5\pi}{4}=\cos\frac{7\pi}{4}$

24. Given Ellipse Equation is $2x^2+y^2-8x-2y+1=0 \Rightarrow 2(x^2-4x)+(y^2-2y)+1=0$

$\Rightarrow 2(x^2-4x+4)-8+(y-1)^2=0 \Rightarrow 2(x-2)^2+(y-1)^2=8 \Rightarrow \frac{(x-2)^2}{4}+\frac{(y-1)^2}{8}=1$

Comparing with Standard Form we have $a^2=4, b^2=8$

$\therefore b^2(1-e^2)=4 \Rightarrow 8(1-e^2)=4 \Rightarrow 1-e^2=\frac{1}{2} \Rightarrow e^2=\frac{1}{2} \Rightarrow e=\frac{1}{\sqrt{2}}$

25. We have the Ellipse equation as $\frac{(x-2)^2}{4}+\frac{(y-1)^2}{8}=1$

$\Rightarrow (\alpha,\beta)=(2,1)$ and a = 2 and b = $2\sqrt{2}$; $\because a<b \Rightarrow$ We have

$a^2=b^2(1-e^2) \Rightarrow$ e = $\frac{1}{\sqrt{2}}$

Foci are $(a,\beta \pm be)=\left(2,\left(1\pm 2\sqrt{2}\times\frac{1}{\sqrt{2}}\right)\right)=(2,1\pm 2)=(2,3)$ and $(2,1)$.

26. Here $a=2, b=2\sqrt{2}$. $\therefore$ Length of Latus Rectum $=\frac{2a^2}{b}=2\sqrt{2}$

27. Given $25x^2+16y^2+100x-64y-236=0$

$25(x^2+4x+4)-100+16 \Rightarrow (y^2-4y+4)-64-236=0$

$25(x+2)^2+16(y-2)^2=100+64+236=400$

$\Rightarrow \frac{(x+2)^2}{16}+\frac{(y-2)^2}{25}=1 \Rightarrow \frac{(x-(-2))^2}{16}+\frac{(y-2)^2}{25}=1$

On comparing with Standard form we have $(\alpha,\beta)=(-2,2)$ and $a^2=16$, $b^2=25$

$\because b^2(1-e^2)=a^2 \Rightarrow 25(1-e^2)=16 \Rightarrow 1-e^2=\frac{16}{25} \Rightarrow e^2=1-\frac{16}{25} \Rightarrow e=\frac{3}{5}$

$\therefore$ Distance between Foci = 2 be = $2\times 5\times\frac{3}{5}=6$

28. a) Given Hyperbola is $\frac{x^2}{2}-\frac{y^2}{1}=1$

$\therefore$ Length of Latus Rectum $=\frac{2b^2}{a}=\frac{32}{3}$

b) Given Hyperbola is $\frac{x^2}{2}-\frac{y^2}{1}=1$, Required product $=\frac{a^2b^2}{a^2+b^2}=\frac{2}{3}$

c) Given hyperbola is $xy=18$,

$\therefore$ Length of Latus Rectum $=2\sqrt{2}c=12$

d) Given $x=\frac{e^t+e^{-t}}{3}=\text{Cosh}\,t, y=\frac{e^t-e^{-t}}{3}=\text{Sinh}\,t \Rightarrow x^2-y^2=1 \ \therefore e=\sqrt{2}$

29. Equation of Chord AB is $\frac{x}{a}\cos\left(\frac{\alpha-\beta}{2}\right)-\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{d+\beta}{2}\right)$

Put $(\pm ae, o)$ in the above equation, we have $\pm\frac{ae}{a}\cos(\alpha-\beta)=\cos\left(\frac{\alpha+\beta}{2}\right)$

$$\Rightarrow \frac{\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}=\pm\frac{e}{1}$$

Applying Componendo and Dividendo $\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{1-e}{1+e}$ or $\frac{1+e}{1-e}$

30. Given that $x^2-y^2=0$ represent a Hyperbola
where $\Delta\neq 0, h^2>ab$. (For Hyperbola $\Delta\neq 0$ and $h^2>ab$), both are correct.

31. Given Equation of Conic is $21x^2-6xy+29y^2+6x-58y-151=0$

$$\Rightarrow 3(x-3y+3)^2+2(3x+y-1)^2=180 \Rightarrow \frac{(x-3y+3)^2}{60}+\frac{(3x+y-1)^2}{90}=1$$

$$\Rightarrow \frac{\left(\frac{x-3y+3}{\sqrt{10}}\right)^2}{(\sqrt{6})^2}+\frac{\left(\frac{3x+y-1}{\sqrt{10}}\right)^2}{3^2}=1$$

$\therefore$ The given conic is the Equation of an Ellipse

We have $a^2=b^2(1-e^2)$

$\Rightarrow 6=9(1-e^2) \Rightarrow (1-e^2)=\frac{6}{9}=\frac{2}{3} \Rightarrow e^2=1-\frac{2}{3} \Rightarrow e^2=\frac{1}{3}$ then $e=\frac{1}{\sqrt{3}}$

32. Length of axes are $2(3)=6$ and $2(\sqrt{6})$

33. Major axes is $x-3y+3=0$ and Minor axes is $3x+y-1=0$

$\therefore$ Centre = Point of Intersection of the above two axis = (0,1).

34. Given $16x^2-3y^2-32x-12y-44=0$
$16(x^2-2x+1)-16-3(y^2+4y+4)+12-44=0$

$$16(x-1)^2-3(y+2)^2=16-12+44=48 \Rightarrow \frac{(x-1)^2}{3}-\frac{(y+2)^2}{16}=1$$

Length of Transverse axis $=2a=2\sqrt{3}$

Length of Conjugate axis $=2b=8$

Centre $C = (1, -2)$, ecentricity = e = $\sqrt{\frac{19}{3}}$

35. Given ellipse $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{16} = 1$

Comparing with Standard Form, we have Centre = $(\alpha, \beta) = (3,2)$ and

$a^2 = 25 \Rightarrow$ a = 5 and $b^2 = 16 \Rightarrow$ b = 4, Hence a > b

Equation of Major Axis is y = 2; Equation of Minor Axis is x = 3

$$e = \sqrt{\frac{a^2 - b^2}{a^2}} = \sqrt{\frac{25 - 16}{25}} = \frac{3}{5}$$

Equation of Directries is $x = \alpha \pm \frac{a}{e} = 3 \pm 5 \times \frac{5}{3} = 3 \pm \frac{25}{3} = \frac{34}{3}, \frac{-16}{3}$

$\Rightarrow 3x = 34, 3x + 16 = 0$

Equation of Latus rectum $x = \alpha \pm ae = 3 \pm \emptyset \times \frac{3}{} = 3 \pm 3 \Rightarrow$ x = 6 and x = 0